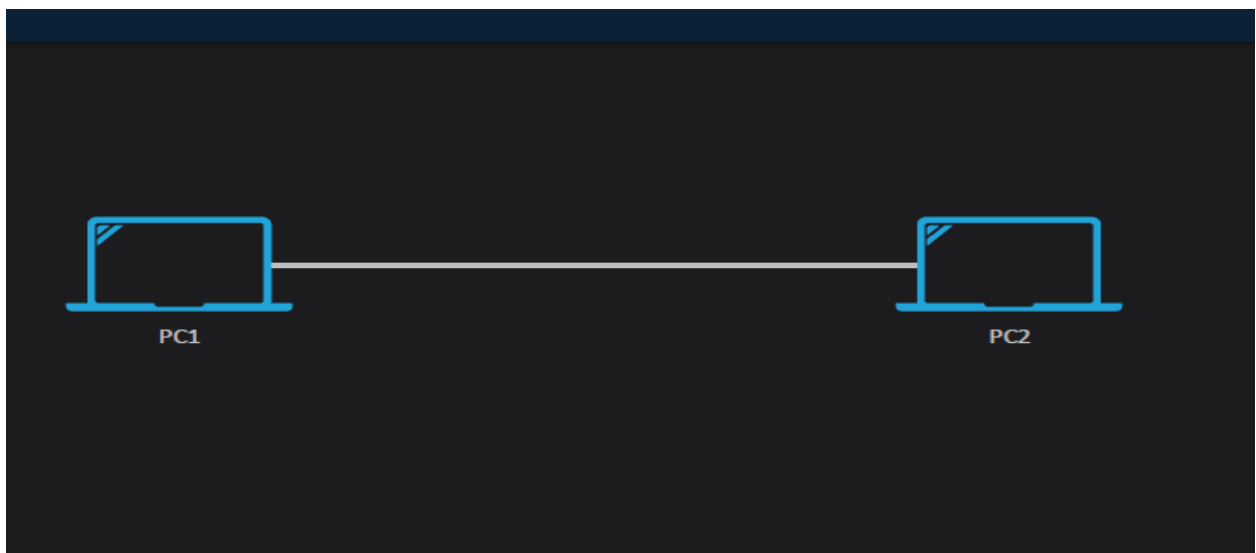


What is a network?

A network is a group of computers which are connected together using wires, optical and radio frequencies to share resources with each other. The resources may include a file or a device such as a printer. These computers can be connected together by using various topologies.

Topology is the way a network is organized. A device in a network is called a node. A node can be anything such as a mobile, laptop or a computer. Each node has its address through which it is identified in the network such as Internet Protocol address (IP address). In a network, any device with an IP address is a host (such as Mobile or Laptop). IP addresses are assigned to devices manually by the administrator or automatically by DHCP (Dynamic Host Configuration Protocol).



What are hubs, switches and bridges?

A hub is a device that broadcasts any incoming data to all devices in the network, without filtering or checking addresses. It receives data from one port and sends it to all the other ports. Only one device at a time can send data at a time. When a hub forwards data, it sends the signal to every device on the network. Only the intended device, whose MAC address matches the destination, accepts the data, while all the other devices simply ignore it. If a person is using a packet sniffer on his device, he can read the data even if it is not for his device. When two devices try to send data together a collision occurs. This usually makes the user use 30% of the speed of the network. **A collision domain** is the point where the data can collide. For example a place in the network such as wire between laptop A and B when they both try to send data on the same time.

A bridge is an old device used to send data to a specific destination using its (destination's) mac address. A bridge has a very few ports (two or three).

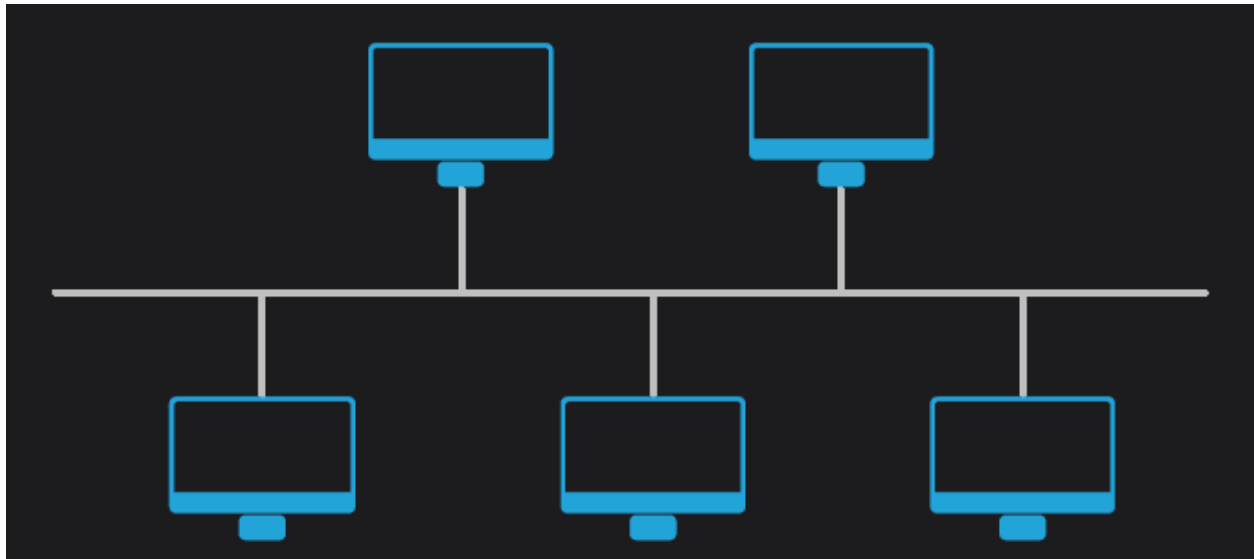
A switch is a modern device that sends data to a specific destination using its mac (destination's) address. It can have multiple ports. It is faster than a bridge.

Feature	Bridge	Switch
OSI Layer	Data Link (Layer 2)	Data Link (Layer 2) → Some can also do Layer 3 (routing)
Ports	Few ports (2–4 typical)	Many ports (dozens to hundreds)
Forwarding method	Software-based (slower, CPU does the work)	Hardware-based using ASIC chips (very fast)
Collision domains	Each port = 1 collision domain, but fewer ports overall	Each port = its own collision domain, scalable
Broadcast domains	One broadcast domain	Still one broadcast domain (unless VLANs are configured)
Speed	Slower, used in early Ethernet days	Much faster, can handle gigabit/10G speeds
Flexibility	Simple, just segments a network	Advanced features: VLANs, STP, port security, QoS, etc.
Usage today	Rarely used	Standard in modern LANs

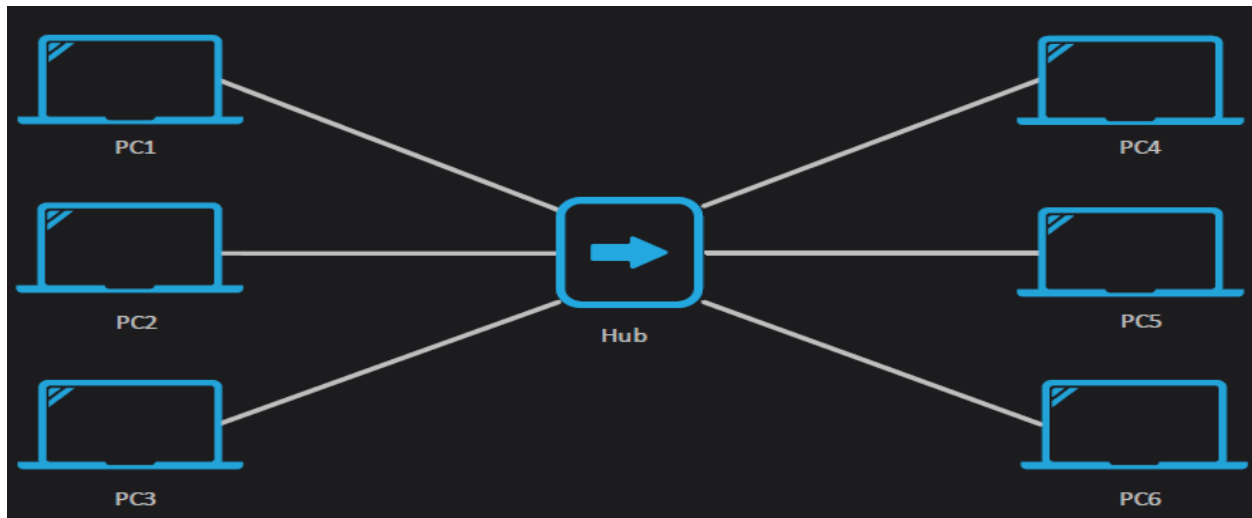
What is a topology?

Topology is the way a network is organized. Each computer in a topology is called a node. There are various topologies such as bus topology, ring topology, star topology etc.

Bus topology is an old and simplest topology. All the devices are connected to a single cable. There are terminators on both ends of the wire so the data/ signals don't bounce back and cause a collision.



Star topology is another type of topology where every node is connected to a hub. Like bus topology, only one node can send data at a time.



What is modulation and demodulation?

Modulation is the process of **changing a carrier signal** (like a wave of electricity or radio) so it can carry information such as voice, video, or data.

Think of it as “packaging” your data onto a wave so it can travel long distances.

Demodulation is the **reverse process of modulation**. It means **extracting the original data** from the carrier signal when it reaches the receiver.

What is a client-server model?

Server: A server is a powerful computer in a network. It takes requests from the other computers of the network and provide them the required services.

Client: A client is a less powerful computer in a network. It is a normal device (computer, laptop, etc) which is used for regular tasks.

Client-server model is a network in which there is one or more servers and there are multiple clients which request the server for services.

What is a Firewall and its types?

A firewall is a device which works as the first line of defense in any network. It stops malicious data from coming and going out of the network. There are different types of firewalls as following:

1. **Ingress (Inbound) Firewalls**
 - Protect the network from **outside threats**.
 - Example: Your home router’s firewall blocks unsolicited traffic from the internet.
2. **Egress (Outbound) Firewalls**
 - Control what **internal devices** can send out to the internet.
 - Used in enterprises to stop data leaks or malware contacting command-and-control servers.

What is IDS and IPS?

An IDS (Intrusion Detection System) is a security system that monitors network traffic, and when it detects something suspicious, it alerts the administrator or forwards the event to a SIEM. It is like a small puppy in a house which will bark if he sees someone breaking into the house at night but will not attack him.

An IPS (Intrusion Prevention System) is a security system that monitors network traffic, and when it detects something suspicious, it actively blocks or prevents the traffic, and

can also alert the administrator or forward the event to a SIEM. It's like a big dog which will bark and attack if someone tries to break into your house.

What is NIC and Mac Address?

A **Network Interface Card (NIC)** is computer hardware that enables a device to connect to a network, either through a cable (wired) or wirelessly (Wi-Fi). It also assigns the device a MAC address for communication.

Mac address is the physical address of a device. It is the address assigned to a device by the manufacturer. Switches and bridges use Mac address to send data. A device such as a laptop can have more than one Mac address. For example if the laptop is connected to the router using a cable, the mac address of laptop will be different from when it is connected to the router with Wi-Fi. Mac address consist of six groups of two hexadecimal digits separated by a hyphens, colons or without a separate.

What is the difference between a Wi-Fi Repeater and Wi-Fi Access Point?

Feature	Repeater	Access Point
Layer	1 (Physical)	2 (Data Link)
Connection to router	Wireless	Wired (Ethernet/fiber)
Creates new Wi-Fi?	No (just extends existing)	Yes (new coverage area)
Speed impact	May reduce speed	Usually maintains full speed
Use case	Quick fix for weak Wi-Fi	Properly expanding a network

What is a LAN and WAN?

A **LAN (Local Area Network)** is a network that connects specific devices, such as computers, smartphones, and printers, within a small location like an office, school, or home. Ethernet and Wi-Fi are two common technologies used in LAN.

A **WAN (Wide Area Network)** connects multiple LANs across large geographic areas, such as between cities or countries, allowing devices in different locations to communicate.

What is Mac table?

A **MAC address table** is a table inside a switch that stores the MAC addresses of devices and maps them to the switch ports they are connected to. This lets the switch forward frames directly to the correct device instead of broadcasting to all. Switches use RAM to store mac tables. Means when the power is off, the mac table gets clear and the switch has to learn the addresses again.

When the mac table is empty and switch has to learn the mac addresses again, the process as following:

- Laptop A sends a packet to Laptop B.
- Now the switch knows the mac address of laptop A by source mac address in the header of the packet which is 1111 (for example) and learn that laptop A is connected to a specific port (port 0/1).
- The switch does not know the mac address of laptop B, so it floods the packet to all the connected devices.
- All the devices except laptop B will ignore the packet and laptop B will keep the packet. By this, switch will learn the mac of laptop A and B and the ports they are connected to.

Why do switches use RAM instead of SSDs?

- Devices connect, disconnect, or move between ports all the time.
- If the switch saved old MAC–port mappings, they’d often be wrong after a restart.
- Example: Yesterday your laptop was on Port 3, today it’s on Port 7. A “saved” table would confuse the switch.

• Learning is fast

- Switches relearn MAC addresses within **milliseconds to a few seconds** after devices send any traffic.
- So, in practice, you hardly notice downtime.

• Efficiency

- Storing and constantly updating millions of MAC entries in permanent storage (like SSDs) would create extra complexity, cost, and wear on storage.
- RAM is much faster for this purpose.

What is ARP and how it works?

ARP (Address Resolution Protocol) is a protocol that allows a computer to discover the MAC address of another device on the same network using its IP address.

Working:

- Computer A wants to send data to Computer B at IP 192.168.1.5.
- Computer A checks its ARP cache (table) — if it doesn’t know the MAC, it sends a **broadcast ARP request**:
“Who has 192.168.1.5? Tell me your MAC.”
- Computer B replies with its MAC address.

- Computer A saves that mapping in its ARP table and uses the MAC to send frames.

Standardization in networks, TCP/IP model and OSI model.

TCP/IP and OSI models were developed to standardize networking, so devices and networks from different vendors could communicate reliably. Before this, proprietary systems made it difficult for networks to interconnect, which benefited vendors but limited users. *{Before them, the communication between two networks was difficult for users but it was good for vendors as they could make money from it (For example before shifting to type C cable, iPhone was using lightning cable while Samsung and google pixel and more were using type C cable, by this iPhone made a lot of money because their phones were only charged by lightening cable)}*

The **OSI (Open Systems Interconnection)** model is a 7-layer conceptual framework that standardizes and explains how different network functions and protocols interact to enable communication between devices. The seven layers of OSI are as following:

1. Physical layer
2. Data-link layer
3. Network layer
4. Transport layer
5. Session layer
6. Presentation Layer
7. Application layer

Please Do Not Throw Pizza Sausage Away (To remember the layers).

The **TCP/IP (Transmission Control Protocol/Internet Protocol)** model is a 4-layer conceptual framework that describes how data is transmitted across the internet and real-world networks using standardized protocols. The 4 layers are as following:

1. Application layer
2. Transport layer
3. Internet layer
4. Network access layer

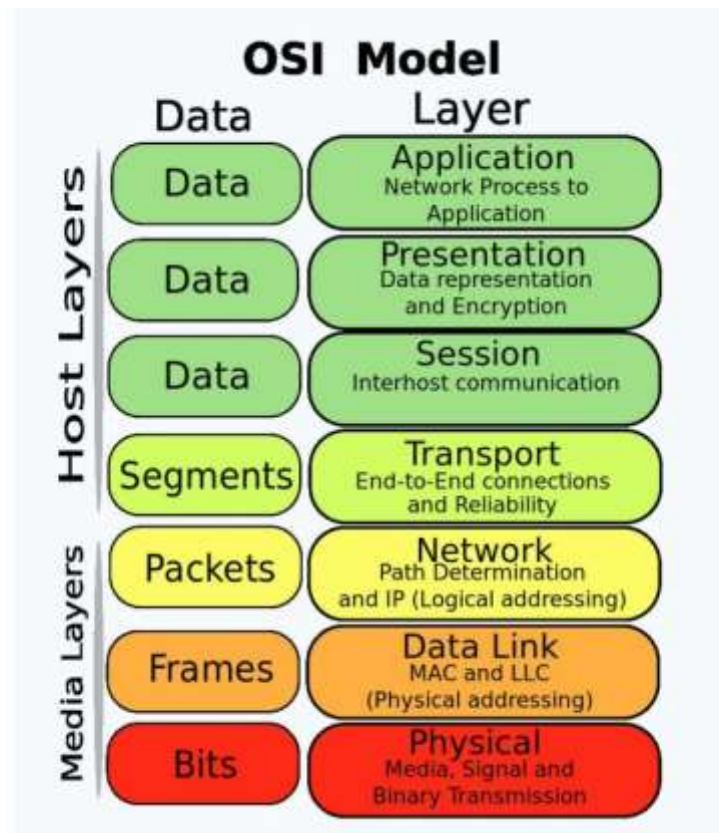
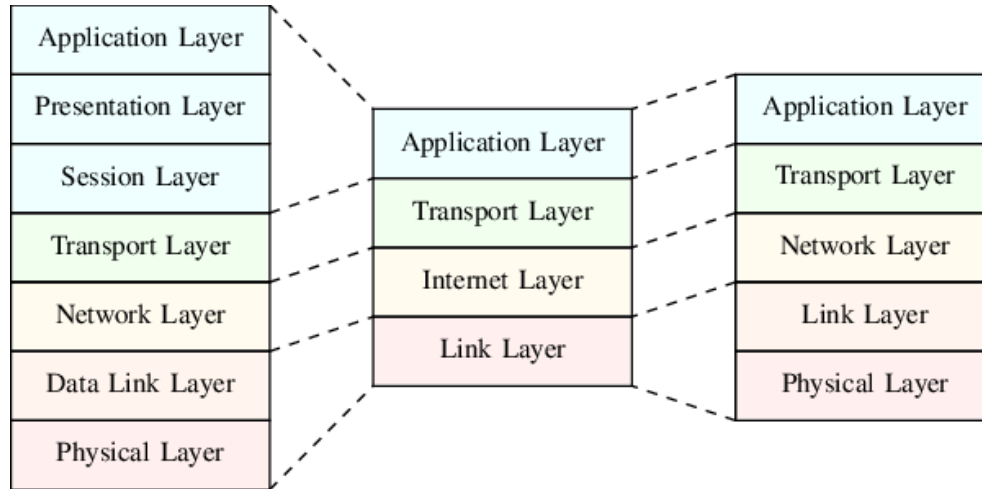
The **TCP/IP Hybrid Model** is a 5-layer conceptual model used to explain networking by combining the clarity of the **OSI model** with the practicality of the **TCP/IP model**.

It is often taught in CCNA and networking courses because:

- It keeps things simple like TCP/IP.
- But it splits out the **physical** and **data link** layers (from OSI) for more clarity.

Layers of it are as following:

1. Application layer
2. Transport layer
3. Internet layer
4. Data-link layer
5. Physical layer



How does port help server to deliver data to right destination?

1. **When Chrome opens a website**
 - Chrome asks the OS: *“Please open a TCP connection to this server.”*
 - The OS assigns an **ephemeral source port** (like 54321).
 - Now the OS has a **socket**: (YourIP, 54321) → (ServerIP, 443).
 - This socket is marked in the OS as *owned by Chrome*.
2. **When the server replies**
 - It sends data back to (YourIP, 54321).
 - The OS checks its socket table:
 - “Ah, port 54321 belongs to Chrome.”
 - The OS delivers that data to Chrome’s process.
3. **Inside Chrome**
 - Chrome takes the data and knows which **tab/session** requested it.
 - (The tab-level mapping is handled by Chrome internally, not by the OS.)

How does data transfer from one layer to another till its destination?

When the data request is sent from Chrome to the server, the first layer it will hit is Transport layer where transport layer will add a header on the request (For example TCP or UDP). Then it will go to the Network layer where a header is added mentioning the IP addresses. Then it will go to link layer where another header with Mac addresses will be added. Then it will go to the physical layer. When the server receives it, the data will go from link layer to network layer to transport layer then application. Each layer till application layer will keep removing the headers, such as link layer will remove mac header and send it to network layer etc.

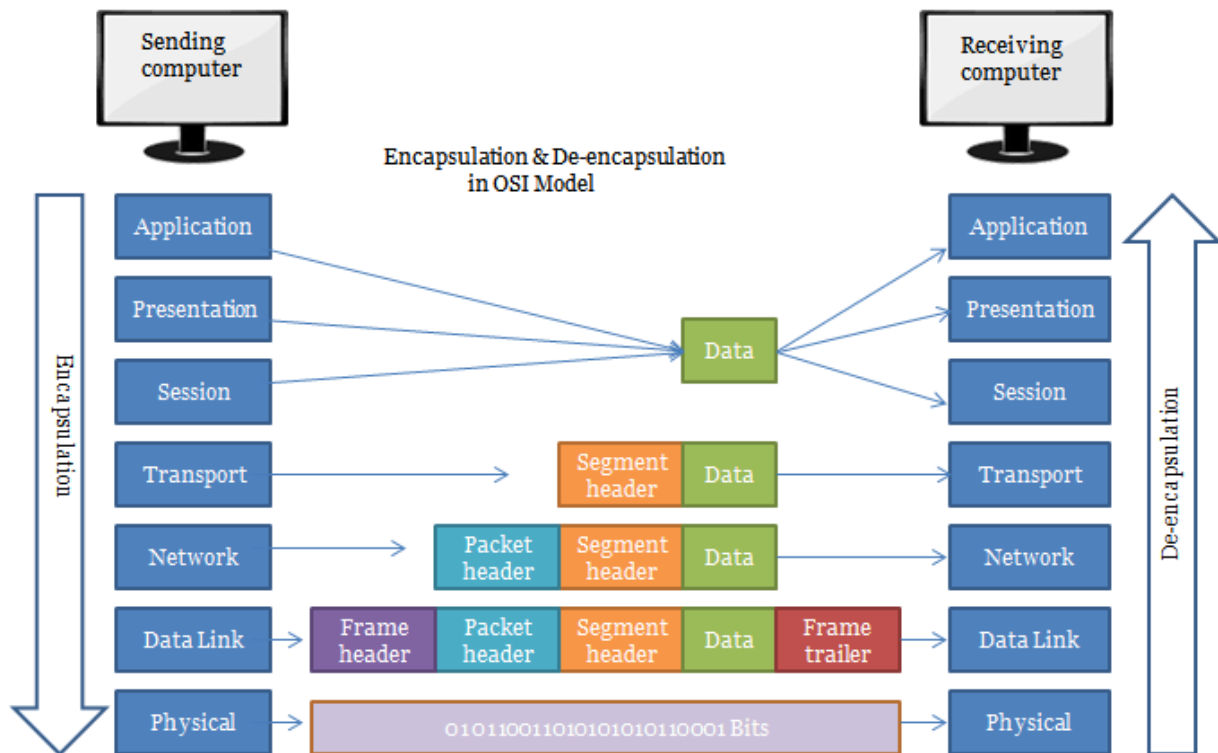
What is Encapsulation and Decapsulation?

Encapsulation (Sender: Your PC → Server)

1. **Application Layer**: Chrome generates an HTTP request (e.g., “GET /index.html”).
2. **Transport Layer**: Adds a **TCP or UDP header** (with source and destination ports).
3. **Network Layer**: Adds an **IP header** (with source and destination IP addresses).
4. **Data Link Layer**: Adds a **MAC header** (with source and destination MAC addresses).
5. **Physical Layer**: Converts the frame into bits (0s and 1s) and sends them as electrical signals, light pulses, or radio waves.

Decapsulation (Server → Receiving the Data)

1. **Physical Layer**: Receives bits and passes them upward.
2. **Data Link Layer**: Reads and removes the **MAC header**, forwards the rest up.
3. **Network Layer**: Reads and removes the **IP header**, forwards up.
4. **Transport Layer**: Reads and removes the **TCP/UDP header**, delivers the payload to the correct port (application).
5. **Application Layer**: The server’s app (like a web server) finally reads the original HTTP request.



What is an IP, IP address, port and socket?

IPv4, Stands for Internet Protocol version 4 is a unique address of a device on the internet, which is used to send the data to the specific device. It is like the house no on a street, each device has its own unique IP address. IP addresses are used in layer three of OSI and 5 layer TCP/IP model. Routers use IP addresses to send data to the destination based on the best route.

Remember, IP (Internet Protocol) is a Protocol, It defines the rules for how data packets are addressed, routed, and delivered across networks. IP address is just a part of its header. **IP is a connectionless protocol.** It means the data is transferred to the other device without building a connection. It doesn't check for errors and guarantee the delivery of data. It is mostly used with a higher level layer (Transport layer) where it use protocols like TCP to establish a connection before delivering the data. But IP alone is used in multiple cases as well. Such as when you ping an IP on the internet/ network, it does not establish a connection and sends data directly. {Ping (ICMP): *Like knocking on someone's door. If they're home, they knock back. But there's no conversation started.*}

A **port** is a 16-bit logical number used by the **Transport layer (TCP/UDP)** to identify a specific application or service on a device. It is not a physical interface like USB or Ethernet port. It is used to tell which app, the data is for. For example you send a message from your Facebook (using chrome tab to another) user. The message will go to the network using its IP and router (in different network) and will find the device using destination mac address. But how

will it identify that the data is for Facebook on receiver's chrome tab 2 and not for some banking application. It is done using port number.

In a client server network (for example a google search), the servers usually use famous ports (for example port 80 for http). The clients usually use random ports which are called **Ephemeral ports**. Ports are assigned to a tab or application by the Operating Systems.

Port	Protocol	Use
20, 21	FTP	File Transfer Protocol (20 = data, 21 = control)
22	SSH	Secure remote access (replaces Telnet)
23	Telnet	Insecure remote access
25	SMTP	Send email
53	DNS	Domain name resolution
67, 68	DHCP	Automatic IP assignment
69	TFTP	Simple file transfers (used for router configs/IOS upgrades)
80	HTTP	Web browsing (insecure)
110	POP3	Retrieve email
143	IMAP	Retrieve email with sync
161/162	SNMP	Device monitoring/management
443	HTTPS	Secure web browsing
514	Syslog	Logging messages from network devices

A **Socket** is a combination of IP address and port number. Here IP address is used to find the device in the network while port number is used by protocols such as TCP or UDP to send data to the specific application.

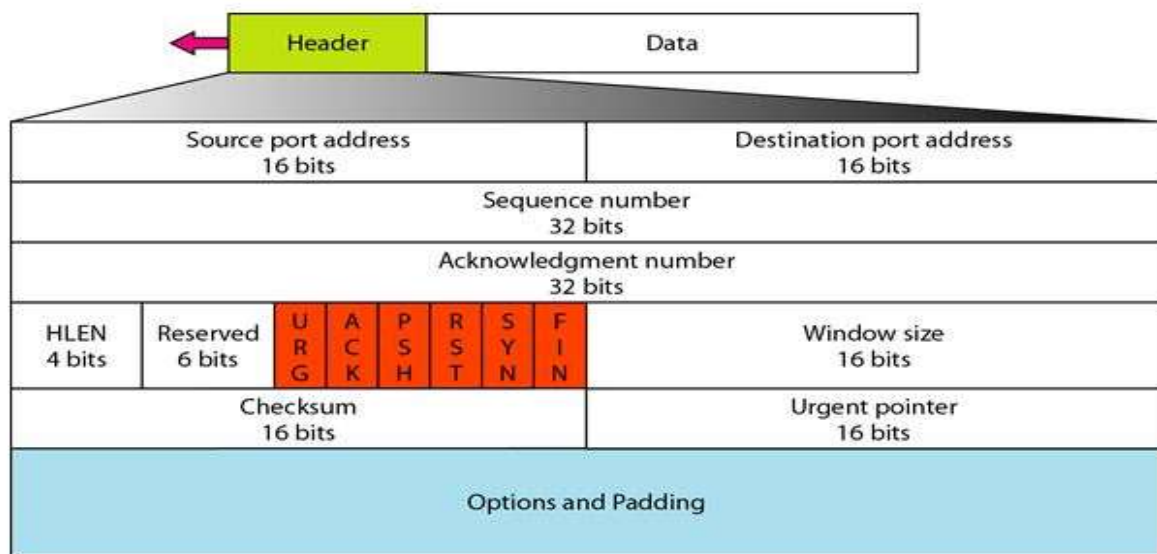
What are the portions of an IP address?

IP address consist of two portions. One portion of it is the network id while the other is the device ID. The network Id is used by the router to find the right network for the data. The device id is used to find the correct device on the network. The **subnet mask** is what tells us where the network portion of the IP ends and where the host portion begins. One network cannot have two devices with same IP addresses (can't have 10.1.1.1 twice on same network) while two devices on different networks can have same IP addresses (such as there can be two devices with same host ID on different networks 11.1.1.1 and 10.1.1.1)

For example when you are going to your friend's house, you will first find the right street (such as Oxford Street), this is how network ID work. And once you are in the street, you will look for the right house (such as house no 1), this is how device id is used. One street cannot have two houses with same house numbers, while two houses can have same house numbers on different streets.

What is TCP and UDP?

TCP (Transmission Control Protocol) is a Layer 4 (Transport Layer) protocol that provides reliable, connection-oriented communication between devices. When a TCP connection is build, the client sends a SYN request to the server. The server replies with SYN ACK and then client sends ACK back to the server. It is also called **Three-way handshake**. TCP make sure the data is delivered and if any packet has lost, it asks the sender to send back the specific packet again. It uses ACK to confirm the data is received. It divide the data into segments and reassemble the data on the receiving end. TCP uses flow control to make sure the sender's data rate matches the receiver's capacity, preventing data loss from buffer overflows. Used in HTTP, HTTPs, FTP, SSH and more.



Upper is the TCP header, it is consist of:

1. Source Port (16 bits)

- Port number of the sender application.
- Example: Your browser (ephemeral port like 49152).

2. Destination Port (16 bits)

- Port number of the receiver application.
- Example: Web server (port 80 for HTTP, 443 for HTTPS).

3. Sequence Number (32 bits)

- Number assigned to the first byte of data in this segment.
- Helps the receiver put packets back in the correct order.
- Example: If the sequence number = 1000 and 500 bytes are sent, the next packet will start with sequence 1500.

4. Acknowledgment Number (32 bits)

- Used to confirm receipt.
- Contains the **next byte the receiver expects**.
- Example: If a segment with seq 1000–1499 is received, the ACK number will be 1500.

5. Header Length (HLEN / Data Offset, 4 bits)

- Tells how long the TCP header is (in 32-bit words).
- Minimum = 20 bytes (no options).
- Needed because TCP header has **variable length** (options may be added).

6. Reserved (6 bits)

- Reserved for future use.
- Always set to 0 in today's TCP.

7. Flags (Control bits, 6 main ones)

These are **1-bit fields** that control the TCP session.

- **URG** → Urgent pointer field significant.
- **ACK** → Acknowledgment number is valid.
- **PSH** → Push function (deliver data to app immediately).
- **RST** → Reset connection.
- **SYN** → Synchronize sequence numbers (used in 3-way handshake).
- **FIN** → Finish (gracefully close connection).

8. Window Size (16 bits)

- Flow control → tells how many bytes the receiver is willing to accept.
- Example: If window = 3000, sender should not send more than 3000 bytes before waiting for ACKs.

9. Checksum (16 bits)

- Error detection, similar to UDP.
- Covers TCP header, data, and a pseudo-header from IP (source IP, dest IP, protocol).

10. Urgent Pointer (16 bits)

- Used with the URG flag.
- Points to the last byte of urgent data in the segment.
- Rarely used today.

11. Options and Padding (variable)

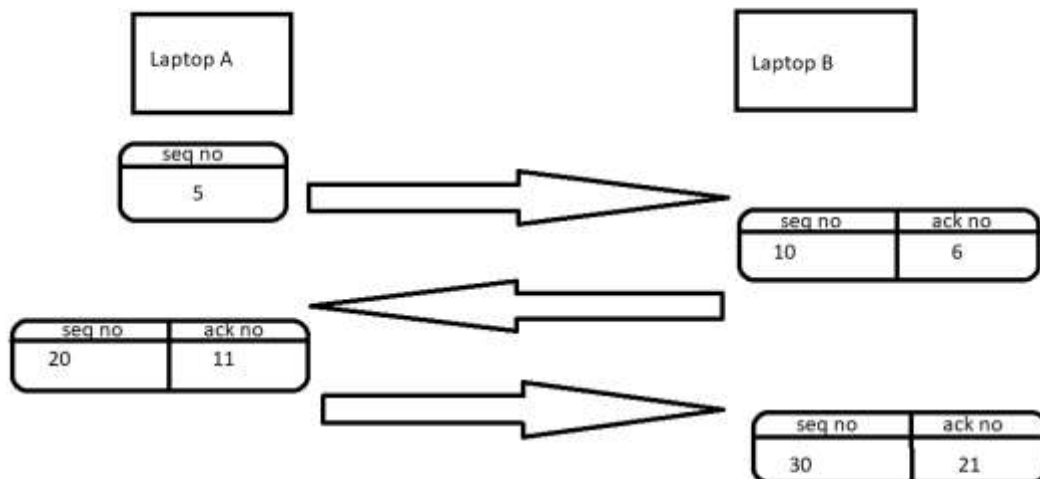
- Used for things like:
 - Maximum Segment Size (MSS)
 - Window Scaling (for high-speed links)
 - Timestamps
- Padding is added to make sure header ends on a 32-bit boundary.

12. Data

- The actual application payload (HTTP request, email, file transfer, etc.).

Three-way handshake:

- In TCP devices build a connection before sharing the data. This connection is called **three-way handshake**.
- Device one sends a SYN message to the device two. This message contains source port, destination port, and sequence no. All other bits of TCP header control flag are set to zero in this except the SYN flag. The sequence no is a random number during the connection building or we can say while establishing the three way handshake.
- The second device receives the message and sends an ACK message to device one. In this TCP header, the acknowledgement number is set to the 1+ of previous sequence number. For example first device sent SEQ no 10, the ACK no from other device will be 11. There will also be a SEQ no in this message. This will also be random. For example the first SEQ no was 10, the other can be 26. In this message, SYN and ACK bit of control flag will be one and all other will be zero.
- In third message of three-way handshake, the first device will send a message where in TCP header, there is a SEQ no, ACK no and the ACK bit will be set to 1 in control flags. SYN bit will be zero which indicated that the connection has established.



- **Window size** is the maximum number of bytes can be sent before getting the ACK from the other device. This effects the throughput of the connection. The higher the window size is, the more the throughput is. For example the window size is 1 and it takes 1 sec for data to traverse from one device to another. Device one sends one byte to device two, and it give an ACK response to device one. To send three bytes of data, it will take 3 seconds. But if the window size is three bytes, device one can send 3 bytes of data and then deceive ACK response from device two. By this, 3 bytes are transferred in one second.
- TCP uses a **sliding window**. It means that the size of window will increase over the time. You may have noticed in real life that when you download a thing from the internet, the downloading speed is relatively slower in the beginning. But over the time, it increases.
- The sender asks the receiver to increase the size of window size over the time. If the receiver can receive that much data, the connection goes on while gradually increasing the window size. If the receiver cannot receive that much data at a time and drops a packet, the congestion window gets half. Over the time, is again increase gradually.
- **Weighted Random early detection (WRED)** is used for quality of service. In a device, packets can be dropped by multiple senders on the same interface. The WRED reduce the window size for each sender so the data doesn't collide. Over the time, it also increases the window size as some senders may slow down while other increase their speed

UDP (User Datagram Protocol) is a Layer 4 (Transport Layer) protocol that provides fast, connectionless communication between devices. It does not make sure if there was any error in data or delivery was correct. It does not use flow control. Used in video streaming platforms such as YouTube or Netflix as few dropped frames doesn't matter but speed does.



Upper is the UDP header, it consists of following things:

- **Source port:** Sender's port number.
- **Destination port:** Receiver's port number.
- **Length:** Length of current packet. For example 8 bits data and 20 bits data, so it will be 28 bits.
- **Checksum:** Checksum in UDP is used to identify if the packet received is correct. It is checked by a hexadecimal number. The sender convert all the info into binary or hexadecimal and then put the calculation in the checksum. The receiver also do the calculation and match the result. If it matches, the receiver keeps the frame otherwise drop it
- **Data:** It is the part where the actual data falls.

What is the difference between UDP and IP?

Feature	UDP (User Datagram Protocol)	IP (Internet Protocol)
OSI Layer	Layer 4 (Transport Layer)	Layer 3 (Network Layer)
Purpose	Delivers data to the correct application on a device (via port numbers).	Delivers data to the correct device across networks (via IP addresses).
Addressing	Uses port numbers (e.g., port 53 for DNS).	Uses IP addresses (e.g., 192.168.1.1).
Connection	Connectionless (no handshake, no session).	Connectionless (forwards packets independently).
Reliability	No reliability (no error correction, retransmission, or ordering).	No reliability (just best-effort delivery).

Feature	UDP (User Datagram Protocol)	IP (Internet Protocol)
Examples of Use	DNS queries, video streaming, gaming, VoIP.	Every packet delivery across the internet depends on IP.

What are IP address classes?

There are 5 IP address classes which are divided into 3 groups as following:

- **Unicast group (Class A, B and C)**
 - **Class A:** First value (from left) in the first octet of class A is always 0. Means the first octet in binary value is from 00000000 to 01111111, in decimal from 0 to 127. But 0 and 127 are already reserved (127 for loopback) so the range of first octet value is from 1 to 126. The first 8 bits in class A are for network while 24 remaining bits are for hosts. Means there can be 2^{24} hosts on class A.
 - **Class B:** The second bit in the first octet of class B is zero, means the first octet is from 10000000 to 10111111, in decimal 128 to 191. The first 16 bits are assigned for the network while last 16 are for Host. There can be 2^{16} hosts in class B.
 - **Class C:** The third bit in the first octet of class C is zero, means the first octet is from 11000000 to 11011111, in decimal 192 to 223. First 24 bits are for the network while last 8 are for hosts. There can be 2^8 hosts in class C.
- **Multicast group (Class D)**
 - **Class D:** The forth bit in the first octet of class C is zero, means the first octet is from 11100000 to 11101111, in decimal 224 to 239. It is used for multicast means the data is sent to multiple devices on same time rather than sending one copy of data at a time. There is no host part in class D IP addresses.
- **Experimental group (Class E)**
 - **Class E:** The fifth bit in the first octet of class C is zero, means the first octet is from 11110000 to 11110111, in decimal 240 to 255. They are reserved for experimental purposes. There are no bits for hosts just like class D.

Class	First Octet Range	Default Mask	Networks	Hosts/Network	Reserved Use
A	1–126	/8	128	16,777,214	127.x.x.x loopback
B	128–191	/16	16,384	65,534	—
C	192–223	/24	2,097,152	254	—

Class	First Octet Range	Default Mask	Networks	Hosts/Network	Reserved Use
D	224–239	N/A	N/A	N/A	Multicast
E	240–255	N/A	N/A	N/A	Experimental

What is a loopback address?

A loopback address is a special IP address which is used to send packets to the same device for checking TCP/IP stack or troubleshooting. **TCP/IP stack** is the set of layers included in TCP/IP model such as Application layer, Transport layer, Internet layer, Link layer. The loopback address in IPv4 is from 127.0.0.1 to 127.255.255.255. It is different for IPv6. The loopback address in IPv6 is ::1. You don't need any connection such as wire or Wi-Fi to loopback. An example of usage of loopback address (usually 127.0.0.1) is when a developer opens his site on his own computer while development. When HTML and CSS pages are combined and developer goes live (in VS code) the site goes live on 127.0.0.1.

The loopback address in the routers or switches is different than host devices. It is also called loopback interface and it can be assigned to a real IP address such as 10.1.1.1. It doesn't go down even if an interface is down. It will only go down if you shut it down by yourself. The router's loopback addresses are also used as the router's ID by the protocols such as EIGRP. If the loopback is not assigned, the protocol will use the highest interface IP as the router ID.

Imagine there are 3 routers, Router 1 is connected to router 2 with two serial cables. The IP of the router 1 and 2 interface is 10.1.1.1, 10.1.1.2, 10.1.1.3 and 10.1.1.4 (for example). And there are two cable connection between router 2 and 3 with IPs from 10.1.1.6 to 10.1.1.9. I assign loopback address to router 1 2 and 3 which are 11.1.1.1, 22.2.2.2 and 33.3.3.3 (for example). Now if you telnet or try to ping router 1 to router 3 on interface 10.1.1.9, it will connect. But if the cable of interface 10.1.1.9 is broken, the telnet or communication will not take place even though there is alternative cable with IP 10.1.1.7 assigned to this interface. But if you telnet 33.3.3.3 from router 1, it will establish a connection no matter on cable is down.

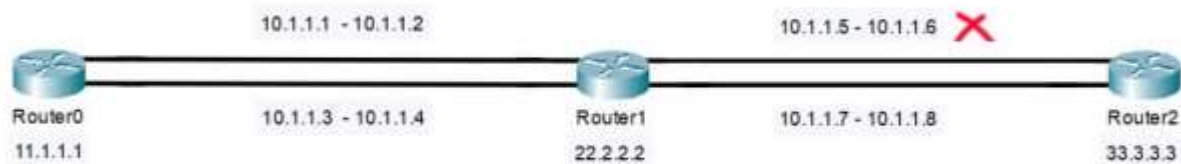
Imagine you have **three routers**:

- **R1 connected to R2** with *two serial links* (IPs: 10.1.1.1–10.1.1.4)
- **R2 connected to R3** with *two serial links* (IPs: 10.1.1.6–10.1.1.9)
- Each router also has a **loopback IP**:
 - R1 → 11.1.1.1
 - R2 → 22.2.2.2
 - R3 → 33.3.3.3

Now, here's the difference:

◆ If I Telnet or ping **R3's interface IP** (say 10.1.1.9), everything works fine. But if that **specific cable goes down**, the connection fails — even though another link (10.1.1.7) is still available.

◆ But if I Telnet or ping **R3's loopback (33.3.3.3)**, the session stays alive. Why? Because the loopback is a **logical, always-up address**. The routing protocol will simply reroute traffic through the alternative cable.



What is RFC1918?

RFC 1918 defines the private IPv4 address ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) that are used in local networks and are not routable on the public internet. They require NAT for external communication. For example your network IP is usually a class C IP address but it will be different than your public IP.

What is local link address?

A link-local address is an IP address valid only within a single local network segment. In IPv6 it always starts with FE80::/10 and is automatically assigned; in IPv4 the APIPA range 169.254.0.0/16 serves a similar role. For example when you connect a PC to another PC using a cable (Ethernet), both the PCs will get a local address just to communicate with each other. The OS assign the PCs the local address.

What is subnet mask, what does subnet do and how it is done?

A **subnet** is a value that is used to find Network ID and Host ID in an IP address. For example the subnet mask of class A will be 255.0.0.0, and class B will be 255.255.0.0 and class C will be 255.255.255.0. Two devices with same subnet can communicate with each other directly without using default gateway / router. But two devices with different subnet masks or network IDs, have to use default gateway in order to communicate with each other. For example if two devices with subnet mask 255.255.0.0 has IP addresses 10.1.1.1 and 10.1.2.1, they can

communicate directly as their network ID is same. But when two devices with subnet mask 255.255.255.0 and IP addresses of 10.1.1.1 and 10.1.2.1 wants to communicate, they have to use a router as their network IDs are different. Subnet mask must have contiguous 1s and 0s such as 11111111.00000000.00000000.00000000 and cannot have discontinuous 1s and 0s such as 10001000.11110011.10101010.00001111.

CIDR (CIDR = Classless Inter Domain Routing) notation is a way of writing subnet masks by appending a slash and the number of network bits to the IP address (e.g., 192.1.1.1/24 means 24 bits for the network, subnet mask 255.255.255.0, and network ID 192.1.1.0).

Formula	Meaning	Why “–2”?
$2^n - 2$	Usable hosts per subnet	Subtract network ID & broadcast
2^n	Number of subnets	All subnet IDs are usable (no subtraction)

Subnetting divides a big network into smaller ones to use IPs efficiently, reduce broadcasts, improve security, and make management easier. For example, in a university, subnetting separates library, faculty, and student networks.

Use of subnetting:

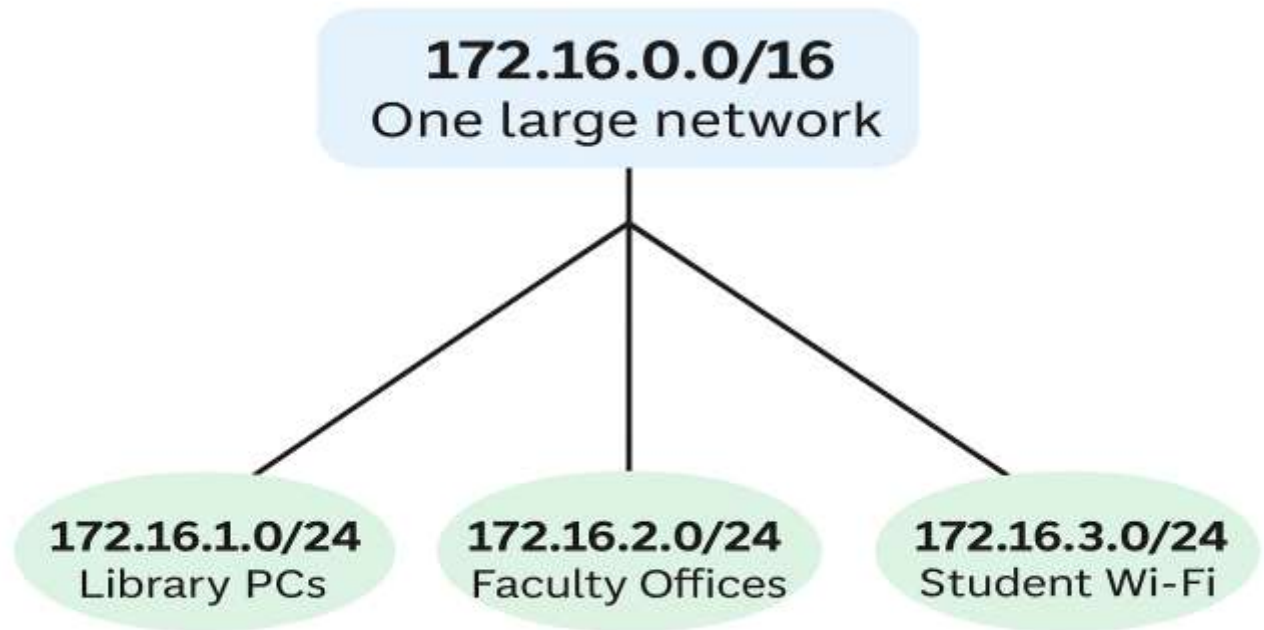
Imagine a **university** with one big Class B network: 172.16.0.0/16.

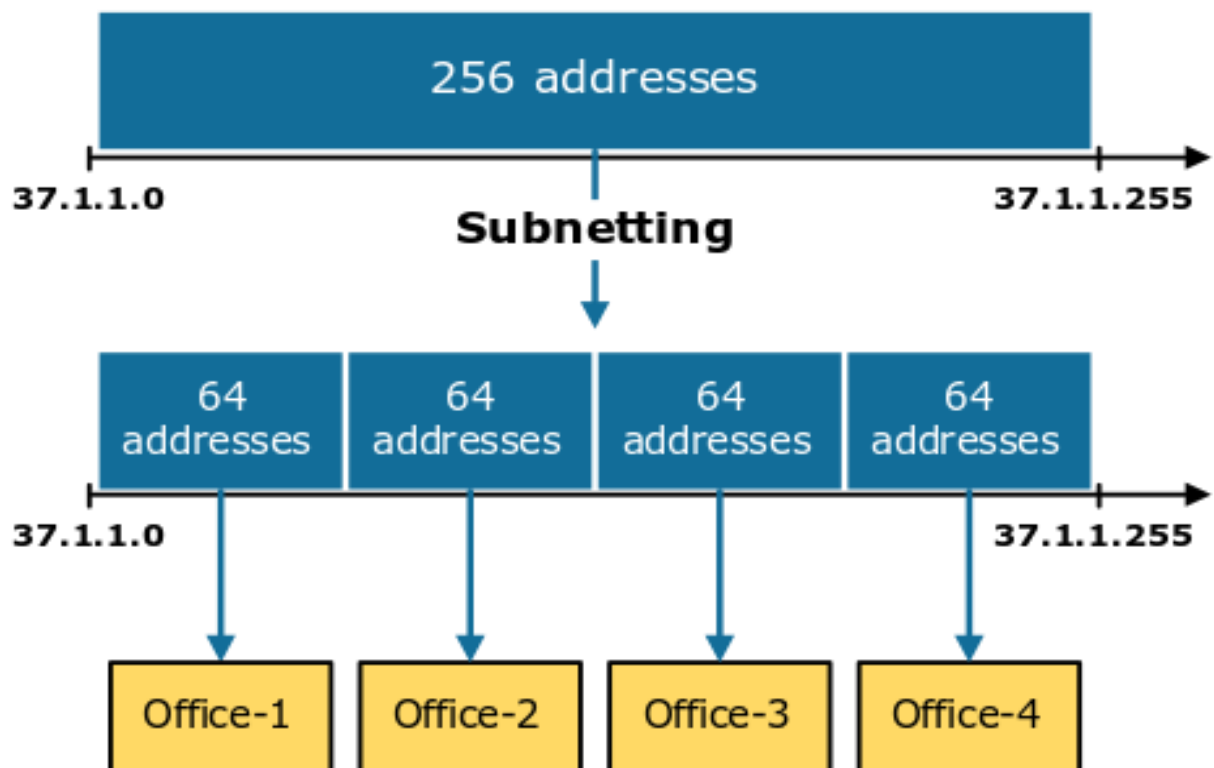
- Without subnetting:
 - All 65,000+ students, staff, servers, Wi-Fi devices are in one giant network.
 - Huge broadcast storms, chaos, no separation.
- With subnetting:
 - Admins divide the /16 into /24 subnets.
 - Example:
 - 172.16.1.0/24 → Library PCs
 - 172.16.2.0/24 → Faculty Offices
 - 172.16.3.0/24 → Student Wi-Fi
 - Each subnet has up to 254 hosts, isolated, easy to manage.

How it works:

- If the subnet mask of IP 172.64.35.10 is 20, it means that the network portion of IP consists of 20 bits from the left.
- So the first octet (172) is a part of network ID because it consists of 8 bits, same for the second octet.

- So the network ID falls in the third octet. Convert the subnet mask into binary.
11111111.11111111.11110000.00000000.
- Here, to see how many hosts, subnet 20 support, we will count the 0s from the right side of the subnet mask, which is 4. That means the subnet /20 allows $2^{12} = 4096$ devices.
- **Block size** = the step between subnet network IDs in the interesting octet. For /20 (mask 240 in 3rd octet), block size is 16, so subnets start at 0,16,32,48... in the 3rd octet.





Two devices cannot communicate to each other directly if their network ID and subnet mask is different. They need a router between them so they can communicate to each other.

For example there are two IP addresses, 192.168.248.1/20 and 192.168.192.1/20. They cannot communicate directly because they are in different subnets. Which we can see better in the binary which is as following:

192.168.248.1/20 in binary:

1100 0000.1010 1000.1111 1000.0000 0001 (ip)

1111 1111.1111 1111.1100 0000.0000 0000 (subnet)

192.168.192.1/20 in binary:

1100 0000.1010 1000.1100 0000.0000 0001 (ip)

1111 1111.1111 1111.1100 0000.0000 0000 (subnet)

Here in the third octet, we can see that both IPs have different network IDs. So to make them communicate without a router, we have to change the subnet mask from 20 to 18, then the third octet will look like:

1111 1000

1100 0000

What is Duplex mismatch, Late collision, Data grams size in a ping, ICMP?

A **duplex mismatch** happens when two connected network devices (like a switch and a computer, or two switches) are set to use different duplex modes:

- **Half-duplex:** Data can only flow in one direction at a time. Devices must take turns sending and receiving.
- **Full-duplex:** Data can flow in both directions simultaneously.

If one side is **full-duplex** and the other is **half-duplex**, they don't communicate properly. The full-duplex device will send and receive freely, while the half-duplex device expects to "wait its turn."

A **late collision** is a special type of collision in Ethernet networks that occurs **after the first 512 bits (64 bytes) of a frame have already been transmitted**.

Why 512 bits matter

- In classic Ethernet (CSMA/CD), collisions are expected and normal — **but only in the first 512 bits**.
- That's the maximum time it takes for a signal to travel across the cable (propagation delay) and return as a collision.
- If a collision happens **after this window**, it's considered *late*, which means something is wrong.

Causes of late collisions

- **Duplex mismatch** (one side half-duplex, the other full-duplex).
- **Network congestion** (too many devices or hubs in a collision domain).
- **Cable problems** (too long, poor quality, or broken cables).
- **Hardware issues** (bad NICs or switch ports).

Why they're bad

- In normal collisions, devices detect the collision early and resend the packet.
- With late collisions, the collision is detected **too late** — so the frame is already corrupted and must be retransmitted. This wastes bandwidth and reduces performance.

Late collision is a problem that usually occurs in old network devices, as modern devices use full duplex, there is very less chances of it happening.

When you run a **ping**, it sends an **ICMP Echo Request** packet to a destination. The **datagram size** (sometimes just called "packet size") is the size of the ICMP payload inside that packet.

How it works

- A ping packet has several parts:
 1. **IP header** → 20 bytes
 2. **ICMP header** → 8 bytes
 3. **ICMP payload (data)** → This is what you control with the **datagram size**

So the **total packet size** = IP header + ICMP header + data payload.

ICMP (Internet Control Message Protocol)

- **Full form:** *Internet Control Message Protocol*
- **Layer:** Works at the **Network layer (Layer 3 of OSI)**, alongside IP.
- **Purpose:** It's used by network devices (routers, hosts) to send **control messages and error reporting** — not to carry user data like TCP/UDP.

When you use ping with size 1000, you're telling the system:

“Put 1000 bytes of data payload inside the ICMP Echo Request.”

But what is inside the data payload?

- It's usually **dummy data (filler)** added just to reach the requested size.
- The content depends on the OS:
 - **Windows:** Fills it with ASCII characters like `abcdefghijklmnopqrstuvw...` repeating.
 - **Linux/Unix:** Fills it with a repeated pattern (commonly `0x61 = a` in hex).
 - **Cisco IOS:** Fills with a simple predictable sequence.

The actual values don't matter — they're just padding. The purpose is to make the packet reach the size you asked for, so you can test MTU, performance, or fragmentation.

What is port spanning?

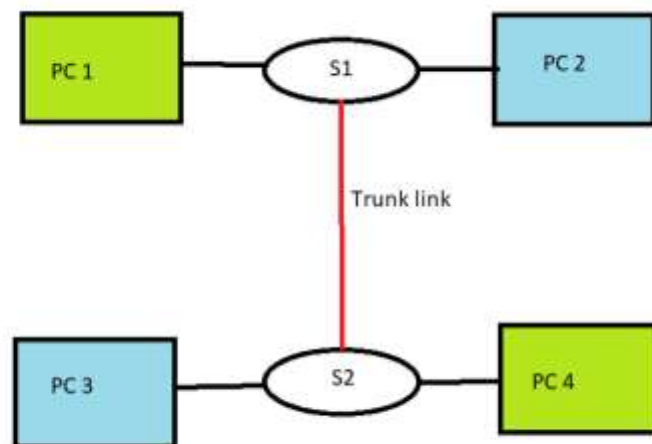
A switch is a smart network device which sends data to specific device after learning the mac addresses. By this, even if you are a part of the network, you cannot analyze the traffic between two devices (if you are in the middle of those devices). For this, you can use **port spanning/ monitoring**. In this, you can monitor traffic of one device on multiple devices. For example you want to monitor the traffic of PC 1 on you PC 3 but there is a switch between both, you can set source interface (in switch) of PC 1 and destination interface (in switch) of PC 3. Now if you use a packet analyzer such as Wireshark, you can analyze the traffic of PC 1 from PC 3.

What is VLAN, trunk link, native VLAN?

VLAN (Virtual LAN) is a virtual network in a switch. If a switch is configured on two VLANs, and two PCs in both VLANs, It will work as two different networks where there are two PCs connected to a single router. Means the logical map for this will be two switches and two pcs per switch. But physically, there is only one switch available and there are for PCs connected with it. Devices on different VLANs cannot communicate to each other without a third layer device such as a router. The PCs in a VLAN does not know if they are on a VLAN or not, they will consider it as a network. Switches use tags in the frames to identify which VLAN does a specific PC belongs to.

A **VLAN trunk link** is a switch port configured to carry traffic from multiple VLANs across a single physical link by tagging frames with VLAN IDs (802.1Q).

- On a trunk link, Ethernet frames are tagged with the VLAN ID (using IEEE 802.1Q standard).
- This tag tells the receiving switch which VLAN the frame belongs to.
- Each switch keeps traffic separated by VLAN, even though they share the same physical link.



In the upper given diagram, there are two switches each connected to one green and one blue VLAN PC. Now PC 1 have to communicate PC 4, he will send data to switch which will send it to switch 2, and switch 2 will send it to green VLAN by reading the VLAN ID. If PC 1 tries to communicate with PC 3, it cannot as both are on different virtual networks, even though they both are using the same cable (trunk link/ red cable) to communicate with their own VLAN.

Native VLAN:

- On an **802.1Q trunk link**, Ethernet frames are usually **tagged** with a VLAN ID (so the receiving switch knows which VLAN they belong to).

- But there's **one VLAN on the trunk that is not tagged** → this is called the **Native VLAN**.

By default, on Cisco switches the **Native VLAN = VLAN 1**.

Why do we need a Native VLAN?

- Some devices (older, non-802.1Q aware devices) cannot understand VLAN tags.
- The Native VLAN allows those untagged frames to still be sent/received on a trunk.
- Basically: **“If no VLAN tag is present, treat it as belonging to the Native VLAN.”**

What is spanning tree?

Spanning Tree Protocol (STP) is a Layer 2 protocol that prevents switching loops in a network with redundant links by logically blocking some ports, ensuring there is always a loop-free path between switches

What will happen without spanning tree protocol:

- Imagine **Switch 1 (S1)** and **Switch 2 (S2)** are connected with **two cables** (two redundant links).
- **PC1 → S1 → PC2** (through S2) wants to communicate.
- Since MAC tables are empty at first, **flooding/broadcasting** happens.
- The frame keeps bouncing back and forth between the two switches over both links (because each switch keeps relearning the MAC address on a different port).
- This creates an **infinite loop = Broadcast Storm** ⚡
- CPU and bandwidth get overwhelmed, eventually crashing the network.

With STP:

- STP detects there are redundant links (potential loops).
- It selects a **root bridge** (main switch) and calculates the best path for traffic.
- Any extra links are put into a **blocking state** (inactive for traffic, but still monitored).

- If the active link fails, STP automatically unblocks the backup link, restoring connectivity.

Root switch Bridge ID and Root port in a topology:

Root switch is the main switch in a topology. It works as the central switch. All the other switches calculate the best path to the root switch. It is selected by STP based on the lowest Bridge ID.

Bridge ID consists of two things. These are bridge priority and MAC address of switch. **Bridge priority** is a value which is initially 32768 and configured by the admin later. If in a topology of three switches, admin has not set bridge priority yet and all switches have same value, STP will make root switch the switch with the lowest MAC address.

Root port is the **single port on a non-root switch** that has the **best path (lowest cost)** to reach the **Root Bridge**. Every non-root switch will have **exactly one Root Port**. The Root Port is always in the **Forwarding state** (it's the way traffic flows *toward* the Root Bridge). For example:

- Suppose **Switch A** is elected as the **Root Bridge**.
- **Switch B** has two links that can reach Switch A. STP will calculate the cost of each path.
 - If Port 1 has a lower cost, Port 1 becomes the **Root Port**.
 - Port 2 might become **Blocked** to avoid loops.

How root port is selected:

1. **Lowest Path Cost**
 - Each port has a cost based on link speed (e.g., Gigabit = 4, FastEthernet = 19, etc. in IEEE default values).
 - The switch sums the cost of the path to the root bridge through each port.
 - The port with the **lowest total cost** to the root wins.
2. **Lowest Neighbor Bridge ID**
 - If two ports have the same cost to the root, the switch looks at the **Bridge ID** (Priority + MAC) of the neighbor switch sending the BPDU.
 - The one coming from the **lower Bridge ID** wins.
3. **Lowest Port ID (Neighbor's Port Priority + Port Number)**
 - If both ports connect to the same switch (same Bridge ID), then it checks the **Port ID** in the received BPDU.
 - Lower Port ID wins.
4. **Local Port Priority / Port Number (Tie-breaker)**
 - If everything else is equal, the switch uses its **own port number/priority** to decide.
 - Lower value wins.

IEEE 802.1D:

It was the spanning tree protocol for bridges. It was released in 1990.

CST (Common Spanning Tree)

- Defined in **IEEE 802.1D**.
- Uses **one STP instance for the entire network**, regardless of the number of VLANs.
- Simple but **not efficient** → if you have 50 VLANs, all of them follow the same spanning tree.
- **Limitation:** Can't optimize traffic per VLAN; not widely used in Cisco environments.
- Means if one interface (or port) is blocked for one VLAN (vlan 10), it will block this port for all other VLANs too.

PVST (Per-VLAN Spanning Tree)

- **Cisco proprietary** (works only on Cisco devices unless both ends support it).
- Runs **one STP instance per VLAN**.
- Advantage: You can choose a **different root bridge per VLAN** → better load balancing.
- **Downside:** More CPU and memory use because it runs separate STP for every VLAN.

What is BPDU

BPDU = Bridge Protocol Data Unit. They are **special control frames** exchanged between switches that are running **Spanning Tree Protocol (STP)**.

Think of them as **“hello/control messages”** switches send to each other to:

- Share their **Bridge ID** (priority + MAC).
- Advertise who they think is the **Root Bridge**.
- Calculate the **shortest path cost** to the Root.
- Decide which ports should be **Root, Designated, or Blocking**.

What are the differences between designated port and root port?

Feature	Root Port (RP)	Designated Port (DP)
Where it exists	On every non-root switch	On every network segment (link)
How many per switch	Only one (best path to root)	Can be multiple (one per segment connected)
Purpose	Forward traffic towards the Root Bridge	Forward traffic away from the Root Bridge
State	Always Forwarding	Always Forwarding

Feature	Root Port (RP)	Designated Port (DP)
Example	Switch → Root Bridge best uplink	On a shared link, the winner that forwards traffic

PortFast:

PortFast is a feature in Spanning Tree Protocol (STP) that allows a port to immediately transition into the forwarding state, skipping the Listening and Learning states. Portfast should only be enabled between end devices, if it is enabled between two switches, it will start looping.

Extended System ID (Newer format – 802.1t / PVST+):

Cisco added VLAN awareness into the Bridge ID.
Now the **16-bit Priority field** is split into two parts:

1. **Priority (4 bits)** → increments in steps of 4096.
2. **Extended System ID (12 bits)** → usually represents the VLAN ID.

So:

- **Bridge ID = 4-bit Priority + 12-bit Extended System ID + 48-bit MAC Address**

How does path cost is calculated?

Bandwidth	Old STP value	New Long STP value
10 Mbps	100	2,000,000
100 Mbps	19	200,000
1 Gbps	4	20,000
N X 1 Gbps	3	10,000
10 Gbps	2	2,000
100 Gbps	N/A	200
1 Tbps	N/A	20
10 Tbps	N/A	2

Path cost is calculated by the speed a port can transfer data on + the number of interfaces (cables) to connect this port to the root switch. For example in a topology S1 is root, S2 is connected to S1 directly and S3 is connected to S1 through S2. So if the ports being used in gigabyte ports, the path cost of S2 to S1 will be 4, and S3 to S1 will be 8 (4 from S3 to S2 + 4 from S2 to S1).

Difference between Rapid PVST+ and PVST:

Feature	PVST	Rapid-PVST+
Protocol	802.1D (STP)	802.1w (RSTP)
Convergence time	30–50 sec	1–10 sec
Port states	5 states (Blocking, Listening, Learning, Forwarding, Disabled)	3 states (Discarding, Learning, Forwarding)
Port roles	Root, Designated, Non-Designated	Root, Designated, Alternate, Backup
Per VLAN instances	Yes	Yes
Load balancing	Yes	Yes
Compatibility	Cisco proprietary	Cisco proprietary, but backward compatible with STP
Use case	Older networks, slower	Modern networks, faster recovery

What is MST?

- **MST** stands for **Multiple Spanning Tree Protocol**.
- It's defined in **IEEE 802.1s** (later folded into 802.1Q).
- It is an **evolution of STP and RSTP** that allows you to run **one STP instance for multiple VLANs** instead of running one per VLAN.

Why MST?

- In **PVST+**, each VLAN runs its **own STP instance**.
 - If you have **100 VLANs**, you run **100 STP processes** → heavy on CPU and memory.
- **MST** groups VLANs into **MST Instances (MSTIs)**.
 - Example: VLANs 1–50 run on MSTI 1, VLANs 51–100 run on MSTI 2.
 - This way, instead of 100 STPs, you just run **2 STPs**.
- Still provides **load balancing** by assigning different VLAN groups to different root switches.

Spanning-tree root guard

In spanning tree if a device is sending false superior BPDUs (and make another device work as root switch without any protocol), it can cause the network getting slow, melt or if there is a hacker behind it, he can see the networks traffic. Root guard is used to make an interface stop getting BPDUs. For example if S1 is getting false BPDUs from g 1/0/4 and you use root guard, it will block that interface. Used On switch-to-switch ports that should never be the root (like access-layer uplinks to distribution).

Loop guard

Loop Guard prevents alternate or root ports from moving to forwarding state if BPDUs are missing.

BPDU guard

Protects against switches (or any device) plugged into **end-user / access ports** that should never see BPDUs. Used on **PortFast ports** (host-facing). It is used to stop blocking BPDU messages from devices such as PC so there is no intentional harm.

What is switch port and routed port?

Switch port is a physical interface on a switch. It uses layer two address (mac address) to communicate with other devices. It can use VLANs and spanning tree protocols.

Routed port is a port on a switch that works as a router's port. It communicate with other devices with layer three address (IP address). A routed port can have IP address while a switch port cannot have it. It does not understand VLANs and spanning trees.

What is ethernetchannel?

Ethernetchannel is a function which is used to group two or more interfaces on a device such as a switch or router. It is used for balancing load between interfaces. For example there are two interfaces between S1 and S2. Because of spanning tree only one will be up at a time while the other will be blocked. If we group both the interfaces, we can use both of them to transfer data simultaneously.

What is VTP and VTY?

VTP (VLAN Trunking Protocol) is a Cisco proprietary protocol used to manage VLANs across a network of switches. Instead of manually creating the same VLANs on every switch, VTP allows one switch (the server) to advertise VLAN information to other switches (the clients) within the same VTP domain. So, when you add, delete, or rename a VLAN on the VTP server, that change automatically propagates to all client switches in the domain. This reduces configuration errors and ensures VLAN consistency.

VTY (Virtual Teletype) lines are virtual terminal lines on a Cisco device. They allow remote access (**via Telnet or SSH**) instead of plugging in with a console cable. Example: If your switch is in a server room, you don't need to stand next to it — you can connect remotely over the network.

What is routed protocol and routing protocol?

Routed protocol is protocol that sends data between two devices. IPv4 and IPv6 are example of routed protocol. They are independent of each other. For example if on two devices IPv4 is not configured right but their IPv6 is configured correctly, they can communicate using IPv6.

Routing protocol is used by the network devices (such as routers) to communicate with other routers to get information about their neighbor network devices and to figure out the best path to deliver the used data sent by using routed protocol. EIGRP, OSPF and RIP are some routing protocols

What is default-gateway and default-route?

When a device (such as PC) in a network does not know the destination address of the data, it sends the data to it (PC's) **default-gateway**. Default gateway is used by the user devices such as mobiles and PCs as well as some layer 3 switches to communicate on layer 3.

When a router in a network does not know the destination address of data, it sends the data to it **default route**. For example if you send a message on an IP address which is not present in the router's routing table, it will send the message to its default route, which in most cases, is the IP address of ISP's router.

What is autonomus system, IGP and EGP?

An **Autonomous System** is a big group of networks and routers under one admin, running one consistent routing policy, identified by an ASN.

IGP (Interior Gateway routing Protocol) is a protocol used between the devices of one autonomous system. For example there are multiple routers in an autonomous system. All of these routers may use protocols like OSPF or EIGRP (if all the network devices are of CISCO) to communicate with each other.

EGP (Exterior Gateway routing Protocol) is a protocol used to connect two autonomous systems. For example two autonomous systems may use BGP protocol to communicate with each other.

What is distance victor protocol and link state protocol?

A **Distance Vector Routing Protocol** calculates the best path based on the **number of hops** (the number of routers a packet passes through) to reach the destination. It does **not consider bandwidth, delay, or other metrics**. These protocols are easy to configure but less efficient in choosing optimal

paths. Imagine a topology of three routers R1, R2 and R3 with all connected to each other like a triangle. The bandwidth between R1 and R3 is 63kbps while the bandwidth between R1, R2 and R2 and R3 is 10 MBps. If R1 wants to send a message to R3, it will use the direct interface between R1 and R3 no matter the bandwidth is so less. RIP is an example of direction vector protocol.

A **Link State Routing Protocol** calculates the best path using a more advanced metric than just hop count. Instead of choosing the shortest hop path, it considers bandwidth, cost, delay, and other factors to determine the lowest cost path. Imagine a topology of three routers R1, R2 and R3 with all connected to each other like a triangle. The bandwidth between R1 and R3 is 63kbps while the bandwidth between R1, R2 and R2 and R3 is 10 MBps. If R1 wants to send a message to R3, it will send the message to R3 through R2.

What is Administrative Distance?

Imagine a router learns about an IP address but OSFP and EIGRP are telling the router two different routes to get to the IP address. The router will determine which path to choose based on the **Administrative Distance** of OSFP and EIGRP. Administrative Distance is a value (0-255) that tells the trustworthiness or reliability of a routing source in a router's routing table. A lower AD means more trust.

Common Administrative Distance values:

Routing Source	AD Value
Connected route	0
Static route	1
EIGRP (summary route)	5
eBGP (external BGP)	20
EIGRP (internal)	90
OSPF	110
IS-IS	115
RIP	120
EIGRP (external)	170
iBGP (internal BGP)	200
Unknown	255 (never trusted)

What is DHCP?

Dynamic Host Configuration Protocol (DHCP) is used to automatically assign IP addresses and other network settings (like subnet mask, default gateway, DNS) to devices in a network.

- This saves time (no need to configure IPs manually).
- Usually, the **router** acts as the DHCP server in small networks, but in larger ones, a **dedicated DHCP server** may be used.

DHCP Process (DORA)

The process goes in **4 steps**:

1. **Discover** – When a device (PC/phone) joins a network, it sends a **DHCP Discover** broadcast asking:
“Is there any DHCP server?”
2. **Offer** – The DHCP server replies with a **DHCP Offer** that includes:
 - An available IP address
 - Subnet mask
 - Default gateway
 - Lease time
3. **Request** – The device responds with a **DHCP Request**, saying:
“Yes, I want to use this IP.”
4. **Acknowledge (ACK)** – The DHCP server sends a **DHCP ACK**, confirming the lease and finalizing the assignment.

What is HSRP?

HSRP (Hot Standby Routing Protocol) is a protocol that allows to router to have same virtual IP address which can be used as a default gateway for other devices. By this, your device will still be able to communicate with other devices in the network even if one router with default gateway is down

What are the main components of a router?

1. **CPU (Central Processing Unit)**
 - The “brain” of the router.
 - Handles instructions, runs the operating system (Cisco IOS), processes routing protocols, and manages overall router functions.
2. **RAM (Random Access Memory)**
 - Temporary, volatile memory (erased when the router is powered off or restarted).
 - Stores:
 - Running configuration file (the active settings).
 - Routing tables.
 - ARP cache.

- Packet buffering (fast switching cache).
- 3. **NVRAM (Non-Volatile RAM)**
 - Stores the **startup configuration** (the settings the router loads when it powers on).
 - Retains data even after the router is powered off.
- 4. **Flash Memory**
 - Non-volatile storage that contains the **Cisco IOS image** (the operating system).
 - Can also store multiple IOS versions, system files, and backup configs.
- 5. **ROM (Read-Only Memory)**
 - Stores **bootstrap program** and basic diagnostics.
 - Helps the router start up and load the IOS from Flash into RAM.
- 6. **Interfaces (Ports)**
 - **Serial Interfaces** → For WAN connections (e.g., connecting two routers).
 - **FastEthernet Ports (100 Mbps)** → For LAN connections.
 - **Gigabit Ethernet Ports (1 Gbps and faster)** → For high-speed LAN/WAN connections.

Difference between Flash memory and NVRAM:

Feature	Flash Memory	NVRAM
Stores	Cisco IOS image (operating system) and sometimes multiple IOS versions/files	Startup configuration file (saved config)
Type	Non-volatile memory	Non-volatile memory
Size	Larger (MBs to GBs)	Smaller (KBs to a few MBs)
Usage at Boot	IOS is copied from Flash into RAM	Startup-config is copied from NVRAM into RAM (becomes running-config)
Editable by User?	Can upgrade/replace IOS images	Can save or erase startup-config (copy running-config startup-config, erase startup-config)
Without It	Router cannot load the IOS → won't function	Router can boot IOS but will start with no configuration (blank/default settings)

When you make a change in the configuration it is saved in RAM, like turning an interface up. To save it to the NVRAM you have to use command `copy running-config startup-config`. Flash memory saves the IOS files of the router.

What is ARP cache?

Arp Cache is a table that stores IPs and their Mac addresses. ARP = Address Resolution Protocol. ARP cache (or ARP table) is a table in the router's or switch's RAM that stores mappings of IP addresses to MAC addresses.

What is packet buffer?

A **packet buffer** is a small section of a router's or switch's RAM used to temporarily store packets when they can't be forwarded immediately, such as when the outgoing interface is busy or slower than the incoming one. It acts like a waiting room that prevents instant packet loss, giving the device time to process or schedule traffic. However, if the buffer becomes full, new packets are dropped (called **tail drop**), which can lead to congestion and performance issues. This is why **QoS (Quality of Service)** is often used to prioritize important traffic in the buffer and reduce the impact of drops.

What is Post Power-on self-test?

Routers and modular switches often support **interface modules or network cards** that can be inserted to expand connectivity, such as a card providing additional FastEthernet, Gigabit Ethernet, or Serial ports. These modules must be installed while the device is **powered off**, because inserting them while powered on can damage the hardware (unless the platform specifically supports *hot-swapping*). When the router or switch is powered on again, it runs the **Power-On Self-Test (POST)**, a diagnostic stored in ROM. During POST, the device checks its hardware components—CPU, RAM, interfaces, and any newly inserted cards—to make sure they are functional and available. Once POST completes successfully, the device continues the boot process: loading the bootstrap program, then the IOS from Flash into RAM, and finally the startup configuration from NVRAM.

What is configuration register?

The **configuration register** is a 16-bit value stored in a Cisco router's NVRAM that controls how the device boots and performs certain low-level functions. It determines whether the router loads the IOS from Flash or another source, and whether it should load the saved startup-config from NVRAM or ignore it. The default value is **0x2102**, which tells the router to boot IOS from Flash and load the startup configuration. A common alternate value is **0x2142**, often used in password recovery, which makes the router ignore the startup-config so it boots without saved settings. You can view the current setting with the `show version` command and change it using `config-register` in global configuration mode. In short, the configuration register acts like the router's BIOS setting, guiding the boot process.

Difference between Configuration register and ROM monitor?

Feature	Configuration Register	ROM Monitor (ROMmon)
What it is	A 16-bit value stored in NVRAM	A low-level mode stored in ROM
Purpose	Controls how the router boots (IOS source, load config or not, console speed, etc.)	Provides a recovery environment when IOS cannot load or when forced
Where stored	NVRAM	ROM

Feature	Configuration Register	ROM Monitor (ROMmon)
Default value	0x2102 (normal boot from Flash, load startup-config)	Not a value — it's a mode you enter into
How to access	By setting the value with <code>config-register</code>	Entered automatically if IOS is missing/corrupt, or manually with <code>Ctrl+Break</code> during boot
Functions	Boot source control, startup-config handling, password recovery (via special values like 0x2142)	Manually boot IOS, load from TFTP/Flash, change config-register, perform password recovery
Analogy (PC)	Like BIOS boot order settings	Like BIOS command shell / recovery mode

What is VTY line?

On Cisco routers and switches, **VTY (Virtual Teletype)** lines define how many simultaneous remote management sessions (Telnet or SSH) a device can accept. For example, `line vty 0 4` enables five VTY lines (0 through 4), meaning the router can handle up to five concurrent incoming remote logins. If all VTY lines are occupied, additional login attempts will be denied until a line is free. The numbering of VTY lines is purely local to the device and does not need to match on the remote router. Importantly, VTY lines limit only incoming connections; a router can initiate outgoing connections (such as ping, Telnet, or SSH to other devices) without restriction, regardless of how many VTY lines it has configured.

Securing a network device

- **Enable password vs enable secret:** *Enable password* sets a plain text password which can be seen in the configuration while *enable secret* sets a hashed password.
- **Closing unused open services:** Disable unused services like `tcp` or `dhcp` on network devices.
- **Closing unused CDP:** CDP is used to share information about neighbors of networking devices. If they are not being used, it's a good practice to close them.
- **Line console and login local:** To set password on a network device like router or switch, we set password or secret in line console 0. If we set login local, we also have to give a username to login.
- **Different VLANs:** It's better to put devices in 2 VLANs and turn the default settings off.

What is a banner in a networking device?

A **banner** is a message displayed on a networking device's terminal session. Banners are often used for legal warnings, informational messages, or notices to users accessing the device. There are various types of banners:

- **MOTD:** It is a temporary banner which settles for a specific time such as 24 hours.
- **Login:** Shows the banner when before logging in.
- **Exec:** It is shown after making a successful login attempt

What is Yersinia?

Yersinia is a kali linux tool to hack layer two and layer three network devices.

What is ARP poisoning?

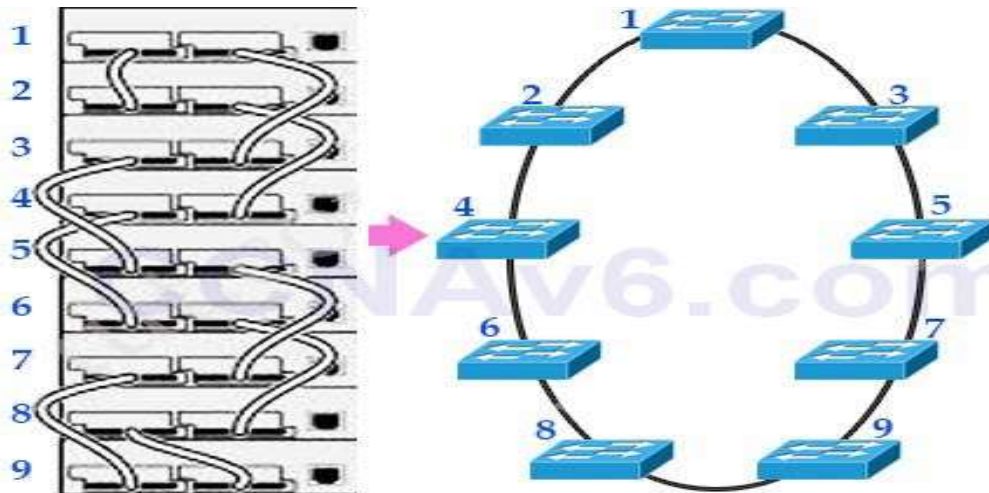
In **ARP poisoning** an attacker forges ARP replies and broadcasts them on the LAN so the ARP payload claims the router's IP (e.g. 192.168.1.1) but uses the attacker's MAC as the sender MAC; the Ethernet frame also comes from the attacker's MAC. When hosts receive these forged ARPs they update their ARP cache to map the gateway IP to the attacker's MAC, so future frames destined for the gateway carry the attacker's MAC as the L2 destination. Because switches forward by MAC, those frames arrive at the attacker's port, allowing the attacker to sniff, modify and forward the traffic to the real gateway (MITM) or simply drop it (DoS). The router still owns its IP, hosts were just tricked into using the wrong MAC, and the attack succeeds because ARP is stateless and trusting. To prevent this, networks use protections like DHCP snooping + Dynamic ARP Inspection (which validate IP-MAC-port bindings), static ARP entries for critical devices, port security, and VLAN segmentation.

What is DHCP snooping?

In **DHCP snooping**, the switch acts as a security guard for DHCP traffic by classifying ports as either trusted or untrusted. Trusted ports (uplinks to legitimate DHCP servers) can send DHCP offers and ACKs, while untrusted ports (access ports to end devices) are restricted to only DHCP requests and replies. If a rogue DHCP server is connected to an untrusted port and tries to hand out IP addresses, its DHCP offers are dropped, preventing clients from being misled with fake IPs, gateways, or DNS settings. At the same time, the switch builds and maintains a DHCP binding table that records the legitimate IP-MAC-port-VLAN associations of clients, which can later be used by features like Dynamic ARP Inspection and IP Source Guard. In this way, DHCP snooping not only stops rogue DHCP servers but also lays the foundation for preventing man-in-the-middle attacks, IP spoofing, and other trust-based abuses on the LAN.

What is switch stacking?

Switch stacking is a technology that allows us to make two (or more switches) work like a single virtual switch. Here, the physical topology is like a ring where each switch is connected to the next while the last one is connected to the first one. All the switches in stacking have shingle telnet, they work like a big switch with various ports. There is a master switch in the stacking which works as the brain of the stacking. It decides which port is best for sending frames.



What is Access Control List (ACL), inbound and outbound ACL, and standard and extended ACL?

An **access list** is a security check applied to traffic on a router interface. It allows or blocks packets based on defined rules and can be used to restrict communication between devices in a network, effectively creating segmentation.

There are **two types of ACLs**:

- An **inbound ACL** checks packets as they enter an interface; if they do not match the rules, the packets are dropped before entering the router.
- An **outbound ACL**, on the other hand, processes packets after they have entered the router and then applies the rules before forwarding them out.

Standard ACLs control traffic based only on the source address, allowing or denying packets from specific devices.

Extended ACLs provide greater control by filtering traffic based on source and destination IP addresses, as well as protocols and services such as HTTP or HTTPS. As a general rule, standard ACLs are best placed near the destination to avoid blocking unintended traffic, while extended ACLs should be placed close to the source to prevent unwanted traffic from traveling through the network.

What is NAT?

In the early days of IPv4, there were enough addresses for all devices. Over time, with the massive growth of the internet, IPv4 addresses became scarce. To solve this, **NAT (Network Address Translation)** was introduced. NAT allows multiple devices inside a private network to share a smaller number of public IP addresses.

When you connect to a home network, your device gets a **private IP address** (visible in your device settings). However, the IP seen on the internet is your **public IP address**, which is usually assigned by your ISP. Private IP addresses never circulate on the public internet — they are translated into a public IP by NAT. For example, whether you use one or ten devices at home, all of them are translated to the same single public IP when accessing the internet.

NAT works with four categories of IP addresses:

- **Inside Local:** the private IP address of your device (e.g., your laptop's 192.168.x.x).
- **Inside Global:** the public IP address representing your device on the internet.
- **Outside Global:** the real public IP address of the destination (e.g., a website server).
- **Outside Local:** how the destination IP is seen inside your network (usually the same as the outside global, unless special NAT is applied).

There are also different types of NAT:

- **Static NAT:** permanently maps one private IP to one public IP.
- **Dynamic NAT:** automatically assigns public IPs from a pool to private IPs on demand.
- **PAT (Port Address Translation):** allows multiple devices to share a single public IP by using different port numbers. This is the most common type used in real-world home and office networks.

What will happen if two devices in a same network use the same port number?

When two devices inside the same network use the same source port and connect to the internet through PAT, the router avoids confusion by changing the source port of at least one of the devices to a different number on the public side. This creates a unique mapping in the NAT table, where each entry is tracked as (Inside Local IP + Port → Inside Global IP + Port). For example, if both PCs used port 50500, the router could translate them to 40001 and 40002 on the public IP. When the destination server replies, the router checks the NAT table and sends each reply back to the correct private device. In short, PAT guarantees uniqueness by rewriting ports so that multiple devices can share a single public IP without mixing up traffic.

What is Quality of Service (QoS)?

Quality of Service (QoS) is used in networks to ensure efficiency and reliability by prioritizing different types of traffic according to their requirements.

- **Data applications**, such as web browsing, file transfers, and email, typically consume more bandwidth but can tolerate packet delay and occasional loss since lost packets can be retransmitted.
- **Voice traffic**, such as VoIP, is highly sensitive to delay, jitter, and packet loss. It requires a minimum guaranteed bandwidth and very low latency because even small delays or dropped packets result in poor call quality, and retransmission is not possible.

- **Video traffic** combines the characteristics of both voice and data. Like voice, it needs consistent bandwidth and low jitter for smooth playback, but it can tolerate slightly more delay because of buffering, and some packet loss without completely breaking the stream.

In summary, voice demands the strictest priority, video requires high but slightly more flexible priority, and data applications are handled with best-effort service since they are more tolerant of delay and loss.

What is Marking and Priority in QoS?

Marking is the process of putting a “label” on packets so the network devices know how to treat them.

- This label is written into specific **header fields**:
 - **Layer 2:** CoS (Class of Service) – 3 bits in the 802.1Q VLAN tag. For this, the link between two devices must be trunk or the tag will not be sent.
 - **Layer 3:** DSCP (Differentiated Services Code Point) – 6 bits in the IP header.
- Marking doesn’t itself give priority — it just **classifies the packet** into a category (voice, video, best-effort, etc.) so routers/switches can queue and forward it accordingly.
- Example: A VoIP phone may mark its traffic with **DSCP EF (Expedited Forwarding, value 46)**.

Priority is the actual treatment a device gives to certain traffic after it has been marked/classified.

- Priority usually means putting that traffic into a **special low-latency queue (LLQ)**, so it gets transmitted first when congestion occurs.
- For example: Voice packets marked as EF are placed into the **priority queue**, while normal data traffic sits in best-effort queues.
- Priority ensures low **delay, jitter, and packet loss** for critical traffic.

What is DSCP?

DSCP (Differentiated Services Code Point) is a 6-bit field in the IP header that is used for Quality of Service (QoS) marking. It is part of the 8-bit Type of Service (ToS) byte in IPv4 (or Traffic Class in IPv6), where the first 6 bits represent the DSCP value and the last 2 bits are reserved for Explicit Congestion Notification (ECN). DSCP values (0–63) are used by routers and switches to classify and prioritize packets, ensuring that critical traffic such as voice (e.g., EF value 46) receives low latency and minimal loss, while normal data traffic is treated as best effort.

Explicit Congestion Notification (ECN) is a 2-bit field in the IP header that works alongside DSCP. It allows routers experiencing congestion to mark packets instead of dropping them. If both endpoints support ECN, the receiver informs the sender about congestion, and the sender reduces its transmission rate. This mechanism helps manage congestion while avoiding packet loss, improving overall efficiency for protocols like TCP.

DSCP values do not work on the principle of “higher bits = faster speed.” Instead, each DSCP value corresponds to a class of service defined by network policies. For example, DSCP 46 (EF) is reserved for voice and mapped to a priority queue with low delay, while DSCP 0 is best effort with no guarantee. The actual priority given to a packet depends on how routers and switches are configured to handle its DSCP marking, not on the numeric size of the DSCP value.

What are the classes such as Assured Forwarding (AF) class?

In the IPv4 header, the Type of Service (ToS) field is 8 bits, of which the first 6 bits are used for DSCP (Differentiated Services Code Point) and the last 2 bits for ECN (Explicit Congestion Notification). Within DSCP, one commonly used marking method is **Assured Forwarding (AF)**, written as AF_{xy}. Here, the first number (x) indicates the forwarding class (1–4), while the second number (y) indicates the drop precedence (1–3). A higher drop precedence means the packet is more likely to be discarded during congestion. For example, AF11 (decimal 10) is Class 1 with low drop probability, AF13 (decimal 14) is Class 1 with high drop probability, and AF41 (decimal 34) is Class 4 with low drop probability. This system allows networks to provide both bandwidth guarantees and differentiated drop behavior under congestion.

<u>Assured Forwarding behavior group</u>				
	Class 1	Class 2	Class 3	Class 4
Low drop probability	AF11 (DSCP 10) 001010	AF21 (DSCP 18) 010010	AF31 (DSCP 26) 011010	AF41 (DSCP 34) 100010
Med drop probability	AF12 (DSCP 12) 001100	AF22 (DSCP 20) 010100	AF32 (DSCP 28) 011100	AF42 (DSCP 36) 100100
High drop probability	AF13 (DSCP 14) 001110	AF23 (DSCP 22) 010110	AF33 (DSCP 30) 011110	AF43 (DSCP 38) 100110

What is IP Precedence and the difference between Class Selector and Assured Forwarding?

IP Precedence (IPP) was the original QoS marking method in IPv4, using the first 3 bits of the ToS field to provide 8 precedence levels (0–7). These levels indicated priority, with 0 as best effort, 5 often used for voice, and 6–7 reserved for network control. Later, the **DSCP (Differentiated Services Code Point)** system expanded this to 6 bits, providing 64 values. To maintain backward compatibility, DSCP introduced **Class Selector (CS) values**, where the first 3 bits represent the old IP Precedence and the remaining 3 bits are set to zero. For example, IP Precedence 5 corresponds to DSCP CS5 (binary 101000, decimal 40).

In DSCP, **Class Selector (CS)** values are designed for backward compatibility with the old IP Precedence markings. CS values use the first three DSCP bits to represent precedence (0–7) while the last three bits are set to zero, e.g., CS2 = binary 010000 (decimal 16). On the other hand, **Assured Forwarding (AF)** provides more granularity by dividing traffic into four classes (AF1–AF4), each with three drop precedences (1–3). For example, AF21 = binary 010010 (decimal 18). While CS values are simple mappings of precedence, AF values allow prioritization within classes and controlled packet drops during congestion. Both CS and AF are subsets of the 6-bit DSCP field.

In IPv4/IPv6, the ToS byte has been redefined into the **DS field** (6-bit DSCP + 2-bit ECN). **Class Selector (CS)** and **Assured Forwarding (AF)** are two different ways of using the 6 DSCP bits. CS values map directly to old IP Precedence levels (first 3 bits = precedence, last 3 bits = 000). AF values, on the other hand, divide traffic into 4 classes and assign 3 drop precedence per class, giving 12 possible DSCP values. They are not separate fields but different DSCP marking schemes.

{If the last 3 bits are all 000, it's a CS value. If not, it could be an AF value (or EF = 101110, etc.)}

Feature	Class Selector (CS)	Assured Forwarding (AF)
Purpose	Backward compatibility with old IP Precedence (IPP)	Modern QoS mechanism for priority + drop precedence
Bits Used	First 3 bits = precedence, last 3 bits = 000	Full 6 bits used (first 3 = class, last 3 = drop precedence)
Range	CS0–CS7 (8 values)	AF11–AF43 (12 values)
Classes	8 precedence levels (0–7)	4 traffic classes (AF1–AF4)
Drop Precedence	Not available (fixed 000)	3 levels (1 = low drop, 3 = high drop)
Example	CS2 = 010000 (decimal 16)	AF21 = 010010 (decimal 18)
Relation	CS maps directly from IP Precedence	AF is independent but can align with CS classes
Use Case	Legacy support, simple classification	Flexible QoS with congestion management

DSCP Code-points:

DSCP (Dec)	Binary	Name
0	000000	CS0 / Default Forwarding
1–7	000001–000111	Reserved
8	001000	CS1
10	001010	AF11
12	001100	AF12
14	001110	AF13
9, 11, 13, 15	(mixed)	Reserved
16	010000	CS2
18	010010	AF21
20	010100	AF22
22	010110	AF23
17, 19, 21, 23	(mixed)	Reserved
24	011000	CS3
26	011010	AF31
28	011100	AF32
30	011110	AF33
25, 27, 29, 31	(mixed)	Reserved
32	100000	CS4
34	100010	AF41
36	100100	AF42
38	100110	AF43
33, 35, 37, 39	(mixed)	Reserved
40	101000	CS5
46	101110	EF (Expedited Forwarding)
41–45, 47	(mixed)	Reserved
48	110000	CS6 (often used for network control)
49–55	110001–110111	Reserved
56	111000	CS7 (highest priority, typically routing/keepalive)
57–63	111001–111111	Reserved

- **CS values** = multiples of 8 (000xxx with last 3 bits = 000).
- **AF values** = in 4 classes (AF1x–AF4x) with drop precedences 1–3.

- **EF** = single value 46 (101110).
- **CS6 & CS7** are usually reserved for **network control / routing protocols**.
- Everything else = reserved or experimental.

QoS Classification Method:

Layer	Field Used	Bits	Examples / Values	Notes
Layer 2 (Data Link)	CoS (Class of Service) in 802.1Q VLAN tag	3 bits (0–7)	CoS 0 = Best Effort, CoS 5 = Voice, CoS 7 = Network Control	Requires a trunk link; tagging only works if 802.1Q is present
Layer 3 (Network)	DSCP (Differentiated Services Code Point) in IP header (redefined ToS field)	6 bits (0–63)	CS0–CS7, AF11–AF43, EF (46)	Most common method; scalable and flexible
	IP Precedence (IPP) (legacy)	3 bits	Values 0–7	Predecessor of DSCP, mapped into Class Selector (CS) values
Application / Transport	Port numbers / NBAR (Network-Based Application Recognition)	TCP/UDP port fields (16 bits)	Port 80 (HTTP), 443 (HTTPS), 5060 (SIP), 5004 (RTP)	Used when classifying traffic based on apps; more granular

What is NBAR?

NBAR is a Cisco IOS feature that allows a router or switch to **identify and classify traffic based on application-layer information**, not just Layer 2 or Layer 3 headers. Normal QoS classification looks at IP addresses, protocols, or port numbers. But many modern applications don't use fixed ports (e.g., P2P, streaming, encrypted apps). NBAR inspects deeper into packet payloads (Layer 4–7) to recognize the application. Once identified, traffic can be placed into QoS classes, policed, or prioritized.

How NBAR works?

- NBAR is **not a protocol** that inserts tags or headers.
- It's a **classification engine** inside Cisco routers/switches.
- When a packet arrives, the device inspects it:
 - First checks **Layer 3 & Layer 4 headers** (IP addresses, ports).

- If that's not enough, NBAR looks **deeper into the payload** (Layer 7) to identify the application signature (like HTTP, Skype, Zoom, etc.).

Once it identifies the traffic, the **router applies QoS marking** (like DSCP, CoS, or queuing policy).

What is Trust Boundary?

A **trust boundary** is the point in a network where you decide **whether to trust or not trust the QoS markings** (like DSCP, IP Precedence, or CoS) on incoming packets.

- End devices (PCs, phones, apps) can set their own QoS markings, but you can't always trust them.
- If you accept markings without verification, a user could **mark all their traffic as high priority** (e.g., set DSCP EF) and abuse the network.
- The **trust boundary is usually placed at the access edge** (e.g., the switch port connected to an IP phone or access point).

Traffic classification criteria in QoS:

Traffic classification is the process of **identifying and categorizing packets** so that QoS policies (marking, queuing, shaping, policing) can be applied. The main **criteria** used are:

Marking-based Classification

- Uses values in **Layer 2 or Layer 3 headers** that indicate priority.
- Examples:
 - **CoS (Class of Service)** → 3-bit field in VLAN tag (Layer 2, IEEE 802.1Q).
 - **DSCP (Differentiated Services Code Point)** → 6-bit field in IPv4/IPv6 header (Layer 3).
 - **IPP (IP Precedence)** → First 3 bits of DSCP field (older method).
- Commonly used because markings are explicit and easy to match.

IP Address-based Classification

- Traffic is classified based on **source/destination IP addresses or subnets**.
- Example: All traffic from 10.1.1.0/24 is treated as "Office LAN traffic."
- Useful for distinguishing departments, sites, or critical servers.

Port/Application-based Classification

- Uses **Layer 4 (TCP/UDP ports)** or deep packet inspection (NBAR) to classify traffic by application.
- Example:
 - Port 80/443 → Web traffic.
 - Port 5060 → SIP signaling (VoIP).

- RTP media streams → Voice traffic.
- NBAR (Network Based Application Recognition) allows classification beyond port numbers, by detecting application signatures.

Protocol-based Classification

- Matches specific **protocols** like ICMP, OSPF, BGP, FTP, etc.
- Example: Prioritizing routing protocol packets (like OSPF hello messages).

Interface/VLAN-based Classification

- Classification based on **where traffic enters the network**.
- Example: All traffic entering on VLAN 10 = "Guest Network" → Apply low priority.
- Often used with **trust boundaries**.

What is Policer and Shaper?

Policers are QoS mechanisms that strictly enforce a traffic rate limit. They monitor the incoming traffic and compare it against a predefined rate, called the Committed Information Rate (CIR). If the traffic exceeds this rate, the policer takes immediate action by either dropping the excess packets or remarking them to a lower priority. This ensures that no user or application can exceed the allocated bandwidth. However, because policers drop traffic rather than buffering it, they can introduce packet loss, which may negatively impact sensitive applications like voice or video. Policers are commonly used by service providers to enforce Service Level Agreements (SLAs) and prevent abuse of bandwidth contracts.

Shapers, on the other hand, are designed to smooth out traffic rather than strictly enforce it. A shaper temporarily buffers packets when traffic exceeds the configured rate, and then releases them steadily so that the overall flow conforms to the allowed bandwidth. Unlike policers, shapers avoid dropping packets unless the buffer becomes full, making them more application-friendly for bursty traffic such as video streaming or file transfers. The trade-off is that shaping introduces extra delay because packets are held in a queue before transmission. Shapers are typically used on egress (outgoing traffic), especially over slower WAN links, to prevent congestion and provide smoother delivery of packets.

{Policers are typically used between routers or at the service provider edge to enforce strict bandwidth limits (ingress). Shapers are commonly used on routers before slower links (egress) to smooth out bursts and prevent congestion.}

What is Queuing and Scheduling?

Queuing is the process of storing and organizing packets in buffers (queues) when there is congestion on an interface (when packets arrive faster than they can be sent out). Packets are classified and placed into different queues based on QoS policy (e.g., voice packets in one queue,

bulk data in another). Without queues, all packets would be treated the same (best-effort), and sensitive traffic like voice/video could suffer delays or drops.

- Examples of queuing mechanisms:

- **FIFO (First In, First Out)** – simple, no priority.
- **PQ (Priority Queuing)** – strict priority for certain traffic (e.g., voice always first).
- **WFQ (Weighted Fair Queuing)** – fair distribution with weights.
- **CBWFQ (Class-Based WFQ)** – queues based on traffic classes.
- **LLQ (Low Latency Queuing)** – guarantees strict priority for real-time traffic + WFQ for others.

Scheduling is the method used to decide which queue packets get sent out first and in what order when multiple queues exist.

- Essentially, queuing creates the separate queues, and scheduling chooses how to serve them.

- Examples of scheduling strategies:

- **Round Robin** – serve each queue in turn.
- **Weighted Round Robin (WRR)** – serve queues based on configured weights (e.g., 70% bandwidth for video, 30% for data).
- **Strict Priority** – always serve the high-priority queue until empty, then move to others.

- Scheduling ensures that high-priority traffic like voice gets low delay while still giving other traffic a fair chance.

What is multi-VLAN access port?

When you connect a PC through an IP phone into a switch, the same switch port is used for two VLANs at once: one for the PC (data VLAN) and one for the phone (voice VLAN). The trick is that the PC does not understand VLAN tags, so it always sends untagged traffic. The switch sees this untagged traffic and places it into the data VLAN, for example VLAN 1. The IP phone, however, is smarter — it adds a small 802.1Q tag on its own voice traffic, so the switch knows this traffic belongs to VLAN 2. In short: **PC traffic = untagged = VLAN 1, Phone traffic = tagged = VLAN 2**. If the PC and the phone ever need to talk to each other, the data cannot go directly between them because they are in different VLANs. Instead, it must go up to the switch and be routed between VLAN 1 and VLAN 2 before coming back down. This is why you can use one physical port for both devices while still keeping their networks separated.

What is CUCM?

CUCM = Cisco's central software that makes an enterprise phone system work, handling **who can call whom, how calls are routed, and what features are available**.

What is SCCP and SIP?

SCCP (Cisco proprietary) is a lightweight signaling protocol where the phone just reports actions to CUCM, and CUCM controls the call. When you press a button on a Cisco IP phone (e.g., extension 1010), the phone sends a simple SCCP message to CUCM like:

SIP (open standard) is more universal and lets different vendors' devices talk to each other. In modern deployments, SIP is more common, while SCCP is mostly used in older Cisco-only setups. When you dial a number on a SIP phone, the phone itself sends a SIP message like:
INVITE sip:1010@company.com

What is OSPF?

OSPF (Open Shortest Path First) is a **dynamic routing protocol** that automatically learns the best path through a network using the **Shortest Path First (SPF)** algorithm by Dijkstra. It quickly adapts to network changes and recalculates routes when links fail.

What are Hello Messages?

Purpose:

Hello messages are used by OSPF routers to discover and maintain neighbor relationships.

Functions:

- Identify and keep track of neighbors
- Ensure routers agree on parameters (Area ID, timers, etc.)
- Maintain active adjacencies

Hello and Dead Intervals

Parameter	Meaning	Default (Cisco)
Hello Interval	Time between Hello packets sent by a router	10 seconds
Dead Interval	Time a router waits without receiving a Hello before declaring the neighbor down	40 seconds

Note:

- Dead Interval = $4 \times$ Hello Interval
- Both intervals **must match** on neighbor routers for adjacency to form.

Example:

- Router sends a Hello every 10 seconds.
- If no Hello is received for 40 seconds → neighbor is marked as *Dead*.

OSPF Packet Types

OSPF uses **five types of packets** to form and maintain neighbor relationships and exchange routing information.

Type	Name	Purpose
1	Hello	Discover and maintain neighbors
2	DB Description (DBD)	Exchange summaries of LSAs (link-state advertisements)
3	Link State Request (LSR)	Request full details of missing LSAs
4	Link State Update (LSU)	Send detailed LSA information
5	Link State Acknowledgment (LSAck)	Confirm receipt of LSAs

OSPF Neighbor Exchange Process

When two OSPF routers form adjacency, they go through these steps:

1. **Hello:** Routers send Hello packets to find and verify neighbors.
2. **Database Description (DBD):** Routers exchange summaries of their link-state databases to see what LSAs each router knows.
Example: "I know routes A, B, C."
3. **Link State Request (LSR):** If one router is missing information, it requests the full details of specific LSAs.
Example: "I need details for route C."
4. **Link State Update (LSU):** Neighbor responds with full LSA information.
Example: "Here are the full details for route C."
5. **Link State Acknowledgment (LSAck):** The receiver acknowledges receipt of the LSA, confirming successful exchange.

What is RID in OSPF?

In OSPF (Open Shortest Path First), every router needs a unique identifier called the **Router ID (RID)**. It is a 32-bit number written in the same format as an IP address, and it is used to identify the router within the OSPF domain. The Router ID does not have to be a reachable IP address; it simply serves as the name of the router in OSPF operations, such as neighbor relationships and link-state advertisements.

How is RID selected?

OSPF chooses the Router ID in a specific order of priority. First, if a Router ID is manually configured using the *router-id* command under the OSPF process that value is always used. If no manual ID is set, OSPF will automatically choose the highest IP address on any loopback interface. Loopbacks are preferred because they are logical interfaces that usually remain up, even if physical interfaces go down. If no loopback interface exists, OSPF will select the highest IP address on any active physical interface. Once the Router ID has been chosen, it does not change automatically even if interfaces go down or new ones come up. To make OSPF recalculate a new Router ID, you must reset the OSPF process using the command *clear ip ospf process*.

What is a designated router, backup designated router and DROTHER?

In OSPF, when routers are connected to a multi-access network (such as Ethernet), many routers share the same broadcast segment. If every router tried to form a full adjacency with every other router on that segment, the number of OSPF connections would increase rapidly, consuming unnecessary bandwidth and CPU resources. To solve this, OSPF uses a system of roles: the Designated Router (DR), the Backup Designated Router (BDR), and DROTHER routers.

The **Designated Router (DR)** acts as the main router on the network segment. It is responsible for collecting and distributing OSPF link-state advertisements (LSAs) between all routers on that segment. The **Backup Designated Router (BDR)** serves as a standby device that automatically takes over the DR's duties if the DR fails. All other routers on the same segment that are neither DR nor BDR are known as **DROTHERs**. These routers do not form full adjacencies with each other; instead, they only form adjacencies with the DR and BDR, which greatly reduces network overhead.

How DR, BDR and DROTHER are elected?

The election of the DR and BDR happens automatically when OSPF routers come online on a broadcast or non-point-to-point network. The selection is based on two main criteria: OSPF interface priority and the Router ID. Each OSPF-enabled interface has a priority value ranging from 0 to 255. The router with the highest priority on that segment becomes the **DR**, and the one with the second highest becomes the **BDR**. If two routers have the same priority, the one with the highest Router ID wins. A router with a priority of **0** is ineligible to become either DR or BDR.

Once the DR and BDR are elected, they maintain their roles until one of them goes down or the OSPF process is manually reset. Even if a new router with a higher priority joins the network later, it will not automatically take over the DR role; OSPF does not preempt existing elections. If the DR fails, the BDR immediately takes over as the new DR, and a new BDR is elected among the remaining routers.

What is IPv6?

IPv6 (Internet Protocol version 6) is the latest version of the Internet Protocol, designed to replace IPv4 and solve its address exhaustion problem. IPv4 could only provide about 4.3 billion unique addresses, which was not enough for the growing number of internet-connected devices. IPv6, on the other hand, uses 128-bit addresses, providing approximately 340 undecillion possible combinations — an unimaginably large number that ensures the world will never run out of IP addresses.

Unlike IPv4, IPv6 does **not require Network Address Translation (NAT)**. In IPv4, NAT was used because public IP addresses were limited, and private networks had to share a single public address to access the internet. With IPv6, every device can have its own **unique globally routable address**, allowing direct end-to-end communication without translation. This simplifies network design, improves performance, and restores the original peer-to-peer nature of the internet.

Each network interface in IPv6 typically has **at least two addresses**. The first is the **Link-Local Address**, which always starts with `fe80::`. It is automatically generated and used only within the local network segment (the same LAN). This address allows communication between devices on the same link, such as between a computer and a router, without requiring internet access. The second is the **Global Unicast Address**, which starts with prefixes like `2000::/3`. This is the address used for communication across the internet when the network's router or ISP provides IPv6 connectivity.

IPv6 addresses can be assigned in several ways: through **SLAAC (Stateless Address Autoconfiguration)**, where a device generates its own address based on router advertisements; through **DHCPv6**, which works similarly to IPv4 DHCP; or through a combination of both. Additionally, IPv6 supports **privacy addresses**, which are temporary random addresses generated for outgoing connections to protect user identity.

Some people assume that giving every device multiple IPv6 addresses wastes space. However, IPv6's design intentionally provides a vast address pool to keep routing and configuration simple. Each local network (LAN) usually receives a **/64 subnet**, which contains about 18 quintillion possible addresses. Even if every device used several IPv6 addresses, it would still consume only a tiny fraction of the total space available. This “waste” is actually a design choice to make IPv6 easy to manage, eliminate NAT, and future-proof the internet for centuries to come.

In summary, IPv6 provides a cleaner, larger, and more efficient addressing system. Every interface automatically gets a link-local address for local use and a global unicast address for worldwide communication. There is no need for NAT, routing becomes simpler, and address configuration can happen automatically. IPv6 is designed not just to solve address exhaustion but to make networking more straightforward, scalable, and secure for the future.

What is EUI IPv6 (IPv6 formatting)?

When IPv6 was first introduced, devices needed a way to automatically generate unique addresses without using DHCP. To achieve this, IPv6 used a method called **EUI-64 (Extended Unique Identifier-64)**. This method takes a device's **48-bit MAC address** and converts it into a **64-bit Interface ID**, which becomes the last part of the IPv6 address. *{FF:FE is inserted based on IEEE's EUI-64 standard.*

It is used to expand a 48-bit MAC address into a 64-bit Interface ID for IPv6.

*The bytes FF:FE were specifically **reserved by IEEE** so they would **never appear in real MAC addresses**, which prevents any **address collisions**.*

Because of this, FF:FE can safely be used in IPv6 address generation and also helps identify that the IPv6 Interface ID was derived from a MAC address.}

Since a MAC address is only 48 bits long, IPv6 expands it to 64 bits by **inserting FF:FE in the middle** and **flipping the 7th bit** (called the *Universal/Local bit*) to mark it as a globally unique identifier. For example, the MAC 98-E7-F4-F5-65-1C becomes 9C-E7-F4-FF-FE-F5-65-1C after conversion. The final IPv6 address might look like 2001:db8:abcd:1234:9ce7:f4ff:fef5:651c.

Although this approach ensures uniqueness, it also creates a **privacy issue** — because the Interface ID contains the device's real MAC address, anyone could track that same device across different networks. To solve this, newer standards introduced **Privacy Extensions (RFC 4941)** and **Stable Privacy Addresses (RFC 7217)**. These methods generate either **temporary random interface IDs** or **consistent non-MAC-based IDs**, so users remain private while still maintaining reliable connectivity.

Today, most modern operating systems like Windows, macOS, and Linux **no longer use EUI-64 by default**. Instead, they rely on these privacy-focused methods to keep IPv6 addresses unique but untraceable.

What is Stateful and Stateless DHCP?

A **Stateful DHCP** is a type of DHCP which assign IP addresses to the clients and then keeps the track of the clients. Such as this IP is assigned to this MAC.

A **Stateless DHCP** is a type in which the server (or router) does not assign IP to the client but it provides the client with the extra info such as DNS-server, DHCP pool, etc. The client generates its IP by itself.

{Note: The concept of stateful and stateless is different only for IPv6. IPv4 only works on stateful DHCP}

What is SLAAC?

In IPv6 networks, **SLAAC (Stateless Address Autoconfiguration)** allows devices to automatically configure their own IP addresses without needing a DHCP server. When a device connects to the network, the router sends a Router Advertisement (RA) that includes the network prefix (like 2001:db8::/64). The device then creates its full IPv6 address by combining that prefix with a unique identifier derived from its hardware, such as its MAC address. This process is called stateless because the router and network do not keep any record of which IPs have been assigned. If the router's advertisement indicates that extra information like DNS servers is available, the device can use stateless DHCPv6 to obtain those details, while still keeping its self-generated address. In contrast, stateful DHCPv6 works like traditional IPv4 DHCP, where the DHCP server assigns and tracks each IP address. Thus, SLAAC makes IPv6 networks more self-managing, scalable, and less dependent on centralized DHCP servers.

What is the difference between OSPFv2 and OSPFv3?

Feature	OSPFv2	OSPFv3
RFC	RFC 2328	RFC 5340
Protocol Type	Link-State Routing Protocol	Link-State Routing Protocol
Used For	IPv4 networks	IPv6 networks
IP Version Supported	IPv4 only	IPv6 only
Router ID (RID)	32-bit (IPv4 format)	32-bit (IPv4 format, still required)
Neighbor Discovery	Uses IPv4 addresses	Uses IPv6 link-local addresses
Interface Identification	Based on IPv4 address	Based on Interface ID , not IP
Network Advertisements	Advertises IPv4 prefixes	Advertises IPv6 prefixes
Authentication	Built into OSPF header (clear-text or MD5)	Removed from OSPF header; uses IPv6 IPsec
LSA Types	Types 1–11	Modified + new LSAs (e.g., Link-LSA, Intra-Area-Prefix-LSA)
Hello Packet Communication	Uses IPv4 unicast/multicast	Uses IPv6 link-local multicast
Multiple Instances per Link	Not supported	Supported
Prefix Handling	Carried within LSAs	Moved to Intra-Area-Prefix LSAs
Authentication Configuration	Configured under OSPF process	Done through IPv6 security (IPsec)

Feature	OSPFv2	OSPFv3
Backward Compatibility	Only for IPv4 OSPFv2 routers	Can coexist but not directly interoperate
Packet Structure	Contains IPv4 fields	Simplified header, supports IPv6
Neighbor Relationship Basis	IPv4 address-based	Interface ID & Link-Local based
Configuration Command (Cisco)	router ospf <id>	ipv6 router ospf <id>
Example Network Command	network 10.1.1.0 0.0.0.255 area 0	interface g0/0 → ipv6 ospf 1 area 0
Security Mechanism	OSPF authentication field	IPv6 IPsec AH/ESP

What is a VPN?

A **VPN** is a *private network established over a public network (like the Internet)* that allows secure communication between users and systems. It protects data when using insecure or clear-text protocols (e.g., HTTP, FTP, Telnet). It also provides privacy, data integrity, and confidentiality.

- **How It Works:**

- VPN encrypts the data before sending it through the public network, making it unreadable to attackers.
- It creates a tunnel between the user and the destination network.

Types of Encryption Used:

1. **Symmetric Encryption:**

- Uses the **same key** for both encryption and decryption.
- Faster but requires secure key distribution.
- Example algorithms: AES, DES.

2. **Asymmetric Encryption:**

- Uses **two different keys** — a public key and a private key.
- Public key encrypts, private key decrypts.
- Used for secure key exchange in VPNs.
- Key sizes usually range from **512 to 2048 bits (or more)**.
- Example algorithms: RSA, Diffie-Hellman.
- Something encrypted by Public key can only be decrypted by the private key and if the data is encrypted via a private key, it can only be decrypted by the public key.
- For example, if A wants to send data to B, It will encrypt the data with B's public key and send it to him. And then the B will decrypt the data.

(Key length (also called key size) is the number of bits used in a cryptographic key. It determines how many possible combinations exist for the key — written as 2^n , where n is the key length in bits. For example $2^8 = 256$)

How does encryption take place?

Secret data -> key -> Encryption -> key -> secret data

Encryption methods:

- **DES** is a symmetric encryption algorithm that uses a 56-bit key and encrypts data in 64-bit blocks. It's now considered insecure due to its short key length.
- **3DES** (Triple Data Encryption Standard) is a symmetric encryption method that enhances the original DES algorithm by applying it three consecutive times on the same data block. It uses either two or three unique keys, which significantly increases its security strength:
 - 112-bit effective key length when the first and third keys are identical (known as two-key 3DES).
 - 168-bit effective key length when all three keys are different (three-key 3DES).
- **AES** is a symmetric encryption standard that supports 128-, 192-, and 256-bit keys, offering strong security and high performance — making it the recommended encryption method for modern use.

What is hashing and why is it used in VPNs?

A **hash function** converts data (like a file or message) into a fixed-length string of characters, called a hash value or message digest. Even a small change in the original data produces a completely different hash, making it easy to detect tampering. Hashing is one-way meaning you can't get the original data back from the hash.

Process of hashing for integrity check:

1. **User (Sender)** prepares the data to send.
2. The sender **encrypts the data** (optional, if confidentiality is also required).
3. The sender then applies a **hash function** (e.g., SHA-256) on the **original or encrypted data** to generate a **hash value**.
4. The **data and the hash value** are both sent to the receiver.
5. The **Receiver** receives the encrypted data and hash value.
6. The receiver then **creates a new hash** from the received data.
7. The receiver **compares** both hash values:
8. If the **hashes match**, the data **has not been altered** (integrity maintained).
9. If the **hashes differ**, the data **was modified or corrupted**.
10. If integrity is confirmed, the receiver can then **decrypt the data** (if it was encrypted).

Authentication of sender:

Authentication is the process of verifying the identity of a user, device, or system before granting access to resources. It ensures that the entity communicating is genuine and not an imposter. Authentication can be achieved through digital signatures or unique identifiers such as router IDs. A **digital signature** is a form of **asymmetrically encrypted data** that proves the sender's authenticity; it is created using the sender's private key and verified with their public key, ensuring authenticity, integrity, and non-repudiation. To share these keys securely, digital certificates are used. Certificates are issued by trusted organizations known as **Certificate Authorities (CAs)** and confirm that a particular public key truly belongs to the individual or organization it represents. For example, if Person A and Person B want to exchange keys securely, they can use a trusted channel with a certificate similar to how a branch manager introduces Person A to Person B, assuring both parties of each other's legitimacy.

What is IPsec?

IPSec (Internet Protocol Security) is a suite of protocols designed to secure IP communications by authenticating and encrypting each IP packet within a communication session. It operates mainly at the network layer (Layer 3) of the OSI model and is commonly used in **VPNs** to ensure secure data transfer across public networks. IPSec provides three main security services: confidentiality, integrity, and authentication. It achieves this through three key components: IKE (Internet Key Exchange), AH (Authentication Header), and ESP (Encapsulating Security Payload).

- **IKE** is responsible for **establishing a secure channel** between two devices by negotiating security parameters and exchanging cryptographic keys automatically.
- **AH** provides data integrity and authentication by verifying that packets have not been altered during transmission, but it **does not provide encryption**.
- **ESP**, on the other hand, offers encryption, authentication, and integrity, ensuring that the data remains private and untampered with.

Modes of IPsec:

IPSec operates in two main modes:

1. **Transport Mode**, only the payload (data part) of the IP packet is encrypted and/or authenticated, while the original IP header remains visible. This mode is typically used for end-to-end communication between two hosts (for example, between a client and a server) where both devices handle encryption and decryption.
2. **Tunnel Mode**, however, the entire IP packet (header + payload) is encrypted and encapsulated within a new IP packet that has a new IP header. This mode is mainly used for network-to-network communication, such as VPNs, where data travels securely between two gateways (routers or firewalls). Tunnel Mode provides an extra layer of protection by hiding the original source and destination IP addresses, making it ideal for site-to-site VPNs.

What are some types of VPNs?

There are three major types of VPNs commonly used for secure communication as following:

1. **Site-to-Site IPSec VPN:** This type of VPN connects entire networks (for example, a company's headquarters and its branch office) through secure tunnels over the internet. The VPN gateways (routers or firewalls) at both sites handle the encryption and decryption, allowing users at both locations to communicate securely without needing individual VPN clients. It operates in IPSec Tunnel Mode, protecting all traffic between the two networks.
2. **Remote Access IPSec VPN:** This VPN allows individual users (like employees working from home) to securely connect to the organization's internal network over the internet. Each user runs VPN client software that establishes an encrypted tunnel to the corporate gateway using IPSec. It provides the same level of encryption as site-to-site VPNs but on a user-to-network basis.
3. **Remote Access SSL VPN:** Instead of IPSec, this VPN uses SSL/TLS encryption the same technology used for securing websites (HTTPS). It allows users to connect securely through a web browser without installing special VPN software. SSL VPNs are ideal for mobile or remote users because they are easier to deploy, cross-platform, and can provide access to specific applications or internal portals rather than the entire network.

What is tunneling in routers?

In networking, **tunneling** is a method used to transport a packet from one network to another by wrapping it inside a different protocol. When you set up a tunnel between two routers, you are essentially creating a "virtual" point-to-point link. Even if those routers are separated by dozens of other devices or the entire internet, they act as if they are directly connected by a single cable.

What is Hypervisor?

A **hypervisor** is the software layer that allows one physical computer to support multiple "guest" virtual machines (VMs). It works by abstracting the physical hardware (CPU, RAM, and Storage) and distributing it among these VMs. This is the foundational technology that makes cloud providers like AWS or Azure possible.

Types of Hypervisor:

Hypervisors are categorized based on where they sit in relation to the physical hardware.

Type 1: Bare Metal

These run **directly on the physical hardware** of the host machine. There is no operating system between the hardware and the hypervisor.

- **Performance:** High, because there is very little overhead.
- **Security:** High, as it's isolated from OS vulnerabilities.
- **Best for:** Enterprise data centers and cloud service providers.

Type 2: Hosted

These run as an **application on top of an existing operating system** (like Windows, macOS, or Linux).

- **Performance:** Lower, because the "guest" VM must go through the "host" OS to reach the hardware.
- **Ease of Use:** High; great for testing and development.
- **Best for:** Personal use, developers, and local lab environments.

What is the difference between Cloud and a Server?

1. The Server: The Individual Machine

A server is a specific piece of hardware (or software) designed to provide a service to other computers (clients).

- **Ownership:** You usually own or rent the specific physical box.
- **Fixed Capacity:** If a server has 32GB of RAM and you need 64GB, you have to physically open the machine and add more.
- **Management:** You are responsible for the hardware, cooling, electricity, and repairs.
- **Location:** Usually sits in an office closet or a dedicated local server room (On-premises).

2. The Cloud: The Infinite Resource Pool

The "Cloud" is a collection of thousands of servers connected together and managed by a provider (like AWS, Azure, or Google Cloud).

- **Abstraction:** You don't see the individual servers; you just see a dashboard where you can ask for "computing power."
- **Elasticity:** You can scale up or down instantly. If your website gets a sudden spike in traffic, the cloud automatically adds more resources.
- **Pay-as-you-go:** You only pay for what you use (by the hour or minute), rather than buying an expensive machine upfront.
- **Redundancy:** If one physical server in a cloud data center fails, your data automatically moves to another one without downtime.

What are different types of radio waves for Wi-Fi?

1. 2.4 GHz: The "Long-Range" Specialist

This band has been the standard for Wi-Fi for decades. Because it uses lower-frequency waves, they are longer and more robust.

- **Range:** Very good. It can travel through walls, floors, and solid objects much better than higher frequencies.
- **Speed:** Slower. It has less bandwidth (fewer "lanes" for data) and is more prone to interference.
- **Congestion:** High. Many household items like microwaves, baby monitors, and Bluetooth devices also use 2.4 GHz, which can cause "traffic jams" in your signal.

2. 5 GHz: The "High-Speed" Specialist

This band was introduced to solve the congestion issues of 2.4 GHz and to provide the high speeds needed for modern streaming and gaming.

- **Range:** Shorter. Higher-frequency waves dissipate faster and struggle to penetrate solid objects like thick brick walls.
- **Speed:** Much faster. It offers significantly more bandwidth and data throughput.
- **Congestion:** Low. It has many more non-overlapping channels, meaning your neighbor's Wi-Fi is much less likely to interfere with yours.

What are Wi-Fi channels for 2.4 GHz?

The 2.4 GHz band is essentially a small strip of radio frequency (from roughly 2.400 GHz to 2.4835 GHz) that is chopped up into smaller segments called **channels**.

There are 14 channels defined in the 2.4 GHz standard, but usage depends on where you live. (North America uses 1-11; most of the world, including Europe and parts of Asia, uses 1-13; Japan uses 1-14). Each channel needs about **20 MHz** (or 22 MHz for older standards) of bandwidth to function. The center frequencies of these channels are only **5 MHz** apart. This math creates the problem: You are trying to fit a 20 MHz "car" into a 5 MHz "parking spot." The only way it works is if the cars park on top of each other.

Non-Overlapping Channels

Because of the heavy overlap, there are only **three** channels that do not touch each other at all in the standard 1-11 range. These are the "Gold Standard" for setting up Wi-Fi networks.

- Channel 1
- Channel 6
- Channel 11

If you have three Wi-Fi routers in a building, you should set one to Ch 1, one to Ch 6, and one to Ch 11. This ensures their waves never cross or interfere with each other.

Overlapping Channels

Any channel that is **not** 1, 6, or 11 is considered an overlapping channel.

For example, if you set your router to **Channel 3**:

- You will interfere with traffic on Channel 1.
- You will **also** interfere with traffic on Channel 6.
- You get the worst of both worlds.

This creates **Adjacent Channel Interference (ACI)**, which is actually worse for performance than just sharing the same channel, because the radios get confused by the partial noise.

What is an Ad hoc network?

An **Ad hoc network** is a decentralized type of wireless network. Unlike a traditional network that relies on existing infrastructure like routers or access points (e.g., your home Wi-Fi), an ad hoc network does not rely on a pre-existing infrastructure. Instead, each device (node) participates in routing by forwarding data for other nodes. The determination of which nodes forward data is made dynamically on the basis of network connectivity and the routing algorithm in use.

How It Works

In a standard Wi-Fi network (Infrastructure Mode), all devices speak to a central Access Point. If "Laptop A" wants to send a file to "Laptop B" sitting next to it, the data goes from A -> Router -> B.

In an **Ad hoc network**:

- "Laptop A" speaks directly to "Laptop B."
- There is no "boss" (router) managing the traffic.
- If devices are too far apart to talk directly, they can pass messages through intermediate devices (multi-hopping), effectively turning every device into a router.

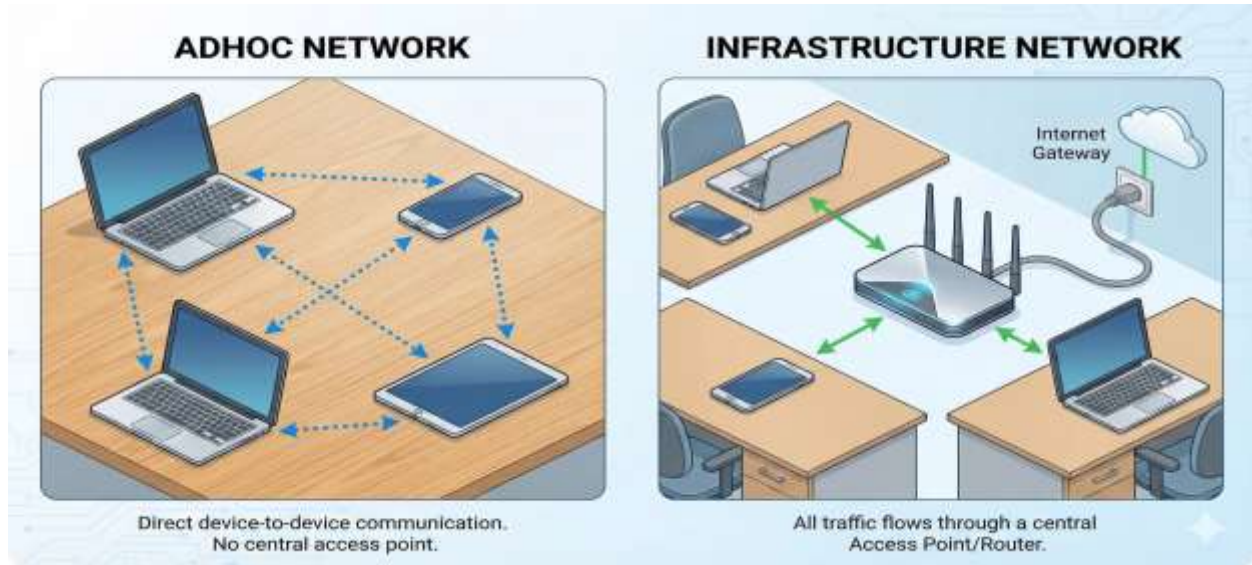
What is an Infrastructure (Wi-Fi) network?

An **Infrastructure Network** is the standard architecture used for most wireless networks today, including your home Wi-Fi and corporate office networks. Unlike an Ad hoc network where devices talk directly to each other, an Infrastructure network uses a central hardware device usually a Wireless Access Point (WAP) or a Router to manage all communication.

How It Works

Think of the Access Point as an air traffic controller.

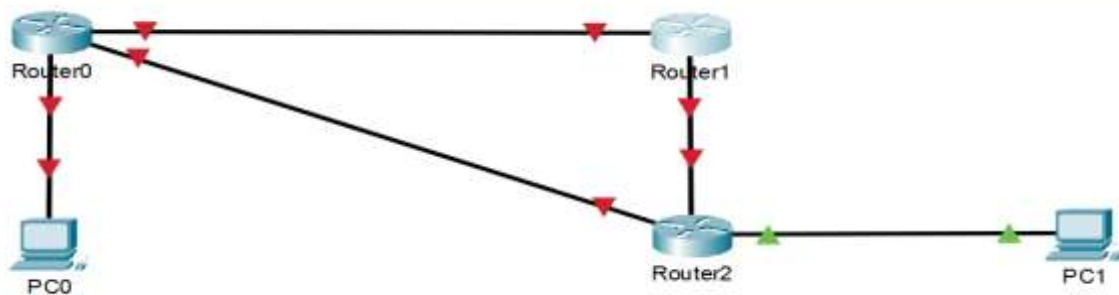
1. **Central Hub:** All devices (stations) connect to the central Access Point (AP).
2. **Traffic Flow:** If "Laptop A" wants to send a file to "Laptop B" sitting right next to it, the data **must** go to the Access Point first, which then forwards it to "Laptop B."
3. **Gateway:** The AP usually connects to a wired network (Ethernet), acting as a bridge to the wider Internet.



What is SDN?

Traditional networks:

In traditional networks, every router makes its own routing decisions, which can result in packets taking different paths depending on their destination.



Planes of a router:

1. **Data plane:** Responsible for forwarding packets from one interface to another.
2. **Control plane:** Responsible for making routing decisions.
3. **Management plane:** Used for configuration, monitoring, and management of the router.



From traditional network to SDN:

- SDN is a network architecture model that centralizes network control
- The control plane is separated from individual routers
- A central SDN controller makes routing decisions

SDN operational overview:

- Install SDN controller on a PC or server
- Use SDN-capable routers and switches
- Devices connect to controller using its IP address
- Network is configured centrally via the controller



{Above we talked about SDN focusing Openflow It was an old technique of SDN where all the devices in the network became dumb and the brain (controller which can be a linux device) takes all the decisions}

What are North and South bounded APIs?

In the Software Defined Networking (SDN) architecture, the Controller is the central "brain." APIs (Application Programming Interfaces) are named based on their direction relative to this controller on a diagram.

- **Southbound APIs:** Go "down" from the Controller to the Network Devices (Switches/Routers).
- **Northbound APIs:** Go "up" from the Controller to the Applications.

1. Southbound APIs

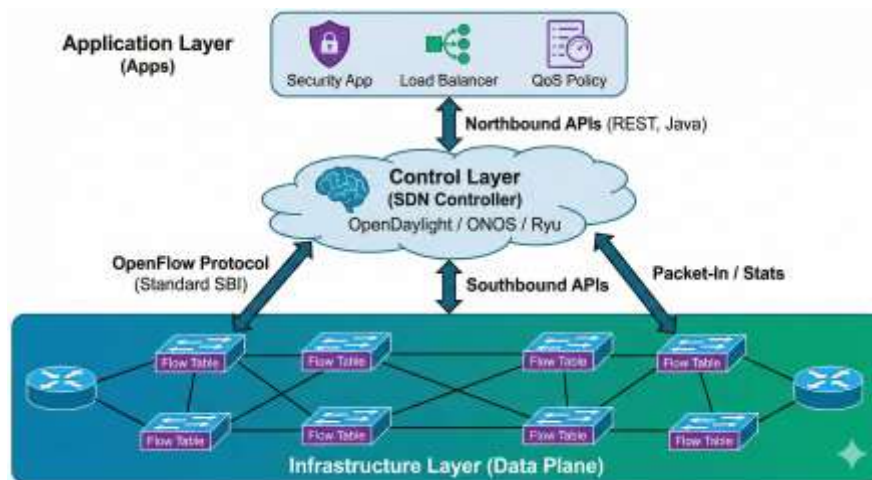
These APIs facilitate communication between the **SDN Controller** and the physical or virtual **network devices** (switches, routers, firewalls).

- **Function:** They allow the controller to push configurations, install flow rules (forwarding instructions), and retrieve network statistics from the hardware.
- **Analogy:** This is the equivalent of the "Device Driver" or the "Assembler" language. It translates high-level decisions into low-level hardware instructions.

2. Northbound APIs (Applications ↔ Controller)

These APIs facilitate communication between the **SDN Controller** and the **business applications** or orchestration systems running on top of the network.

- **Function:** They allow applications to "tell" the network what they need without knowing *how* the hardware works. An app can request "Give me a high-bandwidth path for video conferencing," and the controller handles the complex routing logic via the Southbound API.
- **Analogy:** This is like a high-level programming language (like Python or Java) or a REST API. It abstracts the complexity of the hardware.
- **Standardization Issue:** Unlike Southbound APIs (where OpenFlow is a strong standard), Northbound APIs are **less standardized**. Every Controller vendor (Cisco, Juniper, OpenDaylight) tends to have its own unique Northbound API, usually REST-based.



What is Ansible, Puppet and Chef?

1. Ansible:

Motto: "Simple, Agentless, Powerful."

- **How it works:** Ansible is **Agentless**. You do not need to install any special software on the target servers/routers. It runs on your control machine and connects to the devices using standard **SSH** (secure shell).
- **Language:** Uses **YAML** (Yet Another Markup Language). It is very easy to read (almost like English).
- **Architecture (Push Model):** The centralized server "pushes" the configuration out to the devices. You tell Ansible: "Go configure these 10 servers now."
- **Best for:** Beginners, quick deployments, and network automation (because you can't install agents on most routers).

2. Puppet:

Motto: "Model-Driven and Enforced."

- **How it works:** Puppet is usually **Agent-based**. You must install a piece of software (the Puppet Agent) on every server you want to manage.
- **Language:** Uses its own **Declarative Language** (Puppet DSL). You don't tell it *how* to do something; you tell it *what the end result should look like* (e.g., "Make sure Apache is installed"). Puppet figures out how to do it.
- **Architecture (Pull Model):** The Agents on the servers wake up periodically (e.g., every 30 mins), check in with the Master server, and "pull" down their configuration.
- **Best for:** Large enterprises that need to ensure compliance and consistency over time.

3. Chef:

Motto: "Infrastructure as Code (literally)."

- **How it works:** Like Puppet, Chef is **Agent-based** (Chef Client).
- **Language:** Uses **Ruby** (a programming language). This makes it very powerful but requires you to know how to code. You write "Recipes" which are grouped into "Cookbooks."
- **Architecture (Pull Model):** The Agents check in with the Chef Server to pull down their recipes and execute them.
- **Best for:** Developers and complex environments where you need the full power of a programming language to handle logic and dependencies.

What is Underlay, Overlay and VXLAN?

1. The Underlay (The Physical Foundation)

The **Underlay** is the physical infrastructure of the network.

- **What it is:** The actual switches, routers, cables, and fiber optics.
- **Its Job:** Its *only* goal is to provide reliable IP connectivity from Point A to Point B. It doesn't care about "users" or "applications"; it just moves packets.
- **Protocols:** It uses standard routing protocols like **OSPF**, **IS-IS**, or **BGP** to ensure that every switch can reach every other switch.
- **Analogy:** Think of the Underlay as the **Physical Road System**. It consists of asphalt, traffic lights, and highways. Its job is simply to let vehicles travel.

2. The Overlay (The Virtual Network)

The **Overlay** is a logical (virtual) network created *on top* of the Underlay.

- **What it is:** Tunnels established between devices. Devices in the overlay feel like they are directly connected to each other, even if they are physically separated by 10 routers.
- **Its Job:** To carry user traffic (Virtual Machines, Containers, Tenants). It abstracts the complexity of the physical network.
- **Benefits:** You can create, delete, or move a virtual network in seconds without touching a single physical cable.
- **Analogy:** Think of the Overlay as a **Secure Convoy or VPN** traveling on that road system. The people inside the convoy don't care if the road is asphalt or concrete; they just know they are connected to each other inside their vehicles.

3. VXLAN (Virtual Extensible LAN)

VXLAN is the specific **protocol** used to build the Overlay. It is the "tunneling technology."

- **** The Problem:**** Traditional VLANs (Virtual LANs) have a limit. You can only have **4,096** VLANs in a network. In a massive cloud environment (like Amazon AWS or a large data center), 4,000 segments is not enough.
- **The Solution:** VXLAN extends this limit to **16 Million** segments (IDs).
- **How it works (Encapsulation):**
 1. VXLAN takes the original Layer 2 Ethernet frame (sent by a VM).
 2. It wraps (encapsulates) it inside a Layer 3 UDP packet.
 3. It sends this UDP packet across the Underlay (the physical network).
 4. When it reaches the destination, the wrapper is removed, and the original frame is delivered.

What is SDA?

It is Cisco's specific solution for bringing SDN concepts to the **Campus Network** (your office LAN and Wi-Fi). While standard SDN (like OpenFlow) was academic, and ACI is for Data Centers, SDA is designed for the messy world of user laptops, phones, printers, and IoT devices. In simple terms: **SDA is an automated, policy-based network fabric for the enterprise.**

The Core Idea: The "Fabric"

SDA creates a single, massive **Fabric** across your entire campus.

- **Without SDA:** A user moves from Building A to Building B. Their IP address changes, their VLAN changes, and you have to reconfigure ACLs (firewall rules) on the new switch.
- **With SDA:** The network is one big overlay. The user plugs in anywhere, and the network recognizes them ("Oh, that's You"), keeps their IP address the same, and automatically applies their specific security policies

What is JSON?

JSON stands for **JavaScript Object Notation**.

It is a standardized text format used to store and transport data. Even though it is named after JavaScript, it is **language-independent**. This means Python, Java, C++, and almost every other programming language can read and write JSON.

In the context of **SDN and Networking**, JSON is the primary format used by **Northbound APIs** (REST APIs). When an application talks to an SDN Controller, or when you use Python to get data from a Cisco router, the data usually travels back and forth as JSON.

1. The Structure

JSON is built on two simple structures:

1. **Objects:** Collections of Key/Value pairs. Enclosed in Curly Braces { }.
2. **Arrays:** An ordered list of values. Enclosed in Square Brackets [].

2. A Networking Example

Here is what a JSON response might look like if you asked an SDN Controller for a list of switches.

JSON

```
{
  "network_id": "sdn-campus-01",
  "status": "active",
  "device_count": 2,
  "switches": [
    {
      "hostname": "Switch-A",
      "ip_address": "192.168.1.10",
      "is_active": true,
      "ports": [1, 2, 3, 4]
    },
    {
      "hostname": "Switch-B",
      "ip_address": "192.168.1.11",
      "is_active": false,
      "ports": [1, 2]
    }
  ]
}
```

3. Syntax Rules (The Grammar)

- **Curly Braces { }:** define a container (an object).
- **Square Brackets []:** define a list (an array).
- **Quotes "":** All **Keys** (e.g., "hostname") must be in double quotes. **Strings** (text) must also be in quotes.
- **Colon ::** Separates the Key from the Value.
- **Comma ,:** Separates items in a list or object. (Note: No comma after the last item!).

4. Data Types

JSON supports specific data types as values:

- **String:** "Switch-A"
- **Number:** 10 (No quotes)
- **Boolean:** true or false (Lowercase, no quotes)

- **Null:** `null` (Empty value)
- **Object:** A JSON object inside another JSON object (Nested).
- **Array:** `[1, 2, 3]`

5. JSON vs. YAML vs. XML

Since you asked about Ansible (which uses YAML) and SDN (which uses JSON/XML), here is how they compare:

- **JSON:** Used for **APIs** (Machines talking to Machines). It is strict but easy for computers to parse.
- **YAML:** Used for **Configuration** (Humans talking to Machines, like Ansible). It is cleaner because it uses indentation instead of brackets, making it easier for humans to read.
- **XML:** The "old" standard. It uses tags (like `<hostname>Switch-A</hostname>`). It is heavier and harder to read, so JSON has largely replaced it in modern networking.

What is REST API?

REST API stands for **Representational State Transfer**. In the context of networking and SDN, a REST API is the standard way that different applications talk to each other over the web (HTTP). If **JSON** is the "language" (the text format), then **REST** is the "conversation rules" (how we ask questions and get answers). In your SDN studies, when we talk about **Northbound APIs** (Applications talking to the Controller), they are almost always **REST APIs**.

1. How It Works: The HTTP Verbs

REST APIs use standard HTTP methods (the same ones your web browser uses) to perform actions. These map directly to the **CRUD** (Create, Read, Update, Delete) model used in databases.

HTTP Verb	Action	SQL Equivalent	Networking Example
GET	Read	SELECT	"Show me the configuration of Interface Gi0/1."
POST	Create	INSERT	"Create a new VLAN 10."
PUT / PATCH	Update	UPDATE	"Change the IP address of Interface Gi0/1."

HTTP Verb	Action	SQL Equivalent	Networking Example
DELETE	Delete	DELETE	"Remove VLAN 10."

2. The Components of a Request

When you send a REST request (e.g., using Python or Postman), it has three main parts:

1. **The Endpoint (URL):** The "address" you are calling.
 - *Example:* `https://sdn-controller.com/api/v1/devices`
2. **The Method:** The verb from the table above (GET, POST, etc.).
3. **The Body (Payload):** The data you are sending (usually in **JSON** format).
 - *Note:* You don't need a body for a **GET** request (because you are just asking for info), but you need one for **POST** (to tell the server *what* to create).

3. The Response: Status Codes

When the server answers, it gives you a **Status Code** to tell you if it worked. You likely recognize some of these:

- **200 OK:** Success! Here is your data.
- **201 Created:** Success! I created the resource you asked for.
- **400 Bad Request:** You made a typo in your request (Client Error).
- **401 Unauthorized:** You didn't provide a password/token.
- **403 Forbidden:** You are logged in, but you don't have permission to do this.
- **404 Not Found:** The URL is wrong (The resource doesn't exist).
- **500 Internal Server Error:** The server crashed or had a bug (Server Error).

4. Key Concept: "Stateless"

REST is **Stateless**. This is a critical exam concept. It means the server **does not remember** your previous message. Every single request you send must contain *all* the information needed to complete that request (including your authentication token).

- **Stateful (Analogy):** You walk up to a bartender. "I'll have a coke." Then you say, "Make it two." The bartender remembers the first sentence and gives you two cokes.
- **Stateless (REST Analogy):** You walk up to a vending machine. You put in money and press A1. You get a coke. If you want another, you must put in money and press A1 again. The machine doesn't remember you were just there.

What is DNA center?

Cisco DNA Center (DNAC) is the software **Controller** for Cisco's enterprise networks (Campus and Branch). If you remember our discussion on SDN, we said the "Controller is the brain." **DNA Center is that brain** for office networks (LAN/WLAN). It is the physical appliance (server) that runs the software to design, manage, and troubleshoot the entire network from a single dashboard. It is the heart of what Cisco calls **Intent-Based Networking (IBN)**.

What is the difference between traditional programming and machine learning?

In **Traditional Programming**, *you* write the rules. In **Machine Learning**, the *machine* learns the rules from data.

1. The Core Equation

This is the simplest way to visualize the difference:

- **Traditional Programming:**

$\\$\\$Data + Rules \\rightarrow Answers\\$\\$$

(You give the computer the data and the specific logic, and it gives you the result.)

- **Machine Learning:**

$\\$\\$Data + Answers \\rightarrow Rules\\$\\$$

(You give the computer the data and the desired result, and it figures out the logic/rules to achieve it.)

2. Traditional Programming

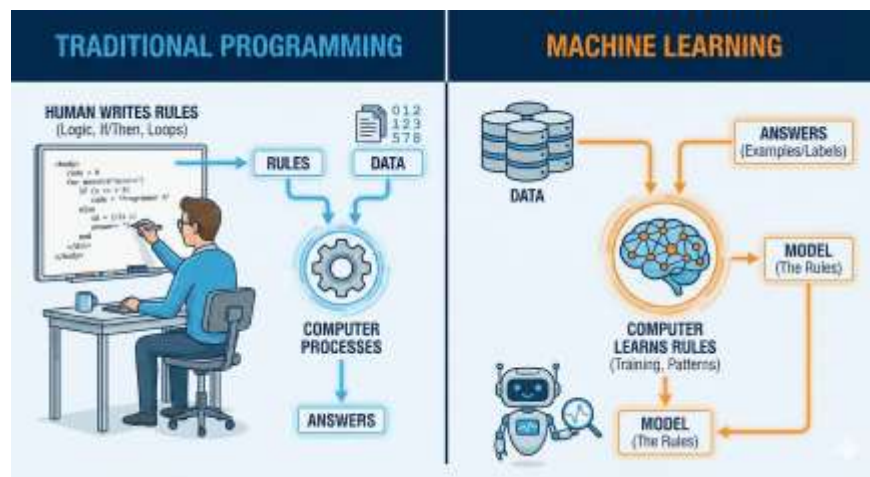
This is what you do when you write a Java or Python script using `if`, `else`, and `for` loops.

- **Process:** You explicitly code the logic. "If the email contains the word 'Winner', mark it as Spam."
- **Limitation:** It is impossible to write rules for everything. How do you write an `if-else` statement to recognize a face? You would need millions of lines of code defining every curve of a nose or shape of an eye. It's too complex.
- **Best For:** Precise, logical tasks (Calculators, Banking systems, Web apps).

3. Machine Learning

This is a subset of AI where you train a model instead of coding it.

- **Process:** You don't write rules. You feed the computer 10,000 emails labeled "Spam" and 10,000 labeled "Not Spam." The algorithm analyzes them and statistically figures out that words like "Winner," "Free," and "Click here" usually mean Spam. It creates its own "rulebook" (the Model).
- **Benefit:** It can solve problems that are too complex for manual coding (Face recognition, Language translation, Recommendation engines).
- **Best For:** Pattern recognition, Prediction, and complex decision making.



What are the types of Machine learning?

1. Supervised Learning (The "Teacher" Approach)

This is the most common type. The model is trained on **labeled data**. This means you show the computer the input *and* the correct answer (the label).

- **How it works:** You give the computer 1,000 emails labeled "Spam" and 1,000 emails labeled "Safe." The computer learns the patterns that distinguish them.
- **Analogy:** A student learning with a teacher. The teacher shows a picture of an apple and says, "This is an apple." Then shows a banana and says, "This is a banana." Later, the student is tested on new fruit.
- **Networking Example: Spam Filters or Firewall Policies.** You teach the firewall: "Packets from this IP are bad" (Label: Threat). "Packets from this IP are good" (Label: Safe).

2. Unsupervised Learning (The "Self-Discovery" Approach)

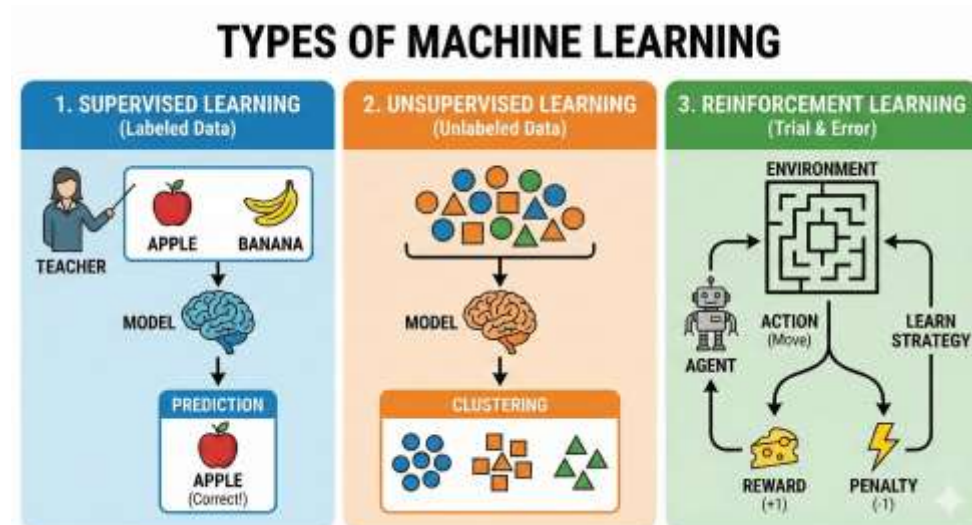
Here, the model is given **unlabeled data**. The computer has no teacher and no answer key. It must explore the data itself to find hidden structures or patterns.

- **How it works:** You dump 1 million server logs into the system. You don't tell it what is "bad." The system analyzes them and says, "Hey, 99% of logs look like *this*, but these 5 logs look weirdly different."
- **Analogy:** A baby playing with toys. No one tells them "This is a block" or "This is a ball," but the baby eventually sorts them into groups based on shape and size (Clustering).
- **Networking Example: Anomaly Detection** (DDoS attacks). The system learns what "normal" traffic looks like. If traffic suddenly spikes or acts weird (zero-day attack), the system flags it, even though it has never seen that specific attack before.

3. Reinforcement Learning (The "Trial and Error" Approach)

This is about learning by interacting with an environment. The agent makes decisions, takes actions, and receives **rewards** or **penalties**.

- **How it works:** A robot tries to walk. If it falls, it gets a negative point (penalty). If it takes a step, it gets a positive point (reward). Over millions of tries, it learns to walk perfectly to maximize its score.
- **Analogy:** Training a dog. You don't explain English to the dog. You just say "Sit." If it sits, you give it a treat (Reward). If it doesn't, you give nothing. Eventually, the dog learns that "Sit" = Treat.
- **Networking Example: Smart Routing** (like in Google Maps or sophisticated SDN). A router tries sending packets via Path A. If they arrive fast, it gets a "reward." If they are lost, it gets a "penalty." Over time, it learns the fastest routes dynamically without a human programming the static route.



What is Predictive and Generative AI?

1. Predictive AI (The Analyst)

Goal: To look at the past and guess the future.

This is the "traditional" AI that businesses have been using for decades. It analyzes historical data to find patterns and make a forecast.

- **How it works:** It takes data in $\rightarrow$ applies statistical algorithms $\rightarrow$ spits out a number or a category.
- **The Output:** A probability, a class, or a forecast.
- **Real-World Examples:**
 - **Netflix/YouTube Recommendations:** "Because you watched X, there is an 85% chance you will like Y."
 - **Weather Forecasting:** "Based on humidity and pressure, there is a 90% chance of rain."
 - **Fraud Detection:** "This credit card transaction doesn't match Yousaf's usual location. Flag it."
 - **Spam Filters:** "This email contains 'free money'. It is 99% likely to be Spam."

2. Generative AI (The Creator)

Goal: To look at the past and create something *new*.

This is the "new wave" of AI (like ChatGPT, Gemini, Midjourney). Instead of just analyzing data, it uses that data to generate entirely original content that never existed before.

- **How it works:** It takes a prompt $\rightarrow$ processes it through massive Neural Networks (LLMs) $\rightarrow$ constructs new data pixel-by-pixel or word-by-word.
- **The Output:** Text, Images, Video, Code, or Audio.
- **Real-World Examples:**
 - **ChatGPT/Gemini:** You ask a question, and it writes a unique essay or poem.
 - **Midjourney/DALL-E:** You describe a "cyberpunk cat," and it paints a picture that no artist has ever painted before.
 - **Coding Assistants:** You write a comment, and the AI writes the actual Python code for you.

